

ECN-BENCH: Design, Systems Engineering, and a Forensic Audit of Multi-Agent Forecasting

Full Technical Report

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Abstract

An LLM that converts a multi-agent transcript into a probability distribution is part of the forecasting system, rather than a neutral scoring function. We study this measurement boundary through a retrospective audit of ECN-BENCH, a social-simulation testbed with four campaigns carrying 30 event identifiers each. On 132 simulated units with identical stored evidence, questions and options, near-uniform forecasts occur in 31 readings by the original extractor and one reading by a replacement. Repeated readings by the replacement also vary: the mean within-unit Brier range is 0.208–0.375 across campaigns. These observations establish sensitivity to the deployed extraction configuration; they do not identify a uniquely correct reading of the swarm. A source audit documents schema bypass and silent failure handling, but changing the evaluator also changes other configuration details. An existing truncation and enforcement experiment on a proxy model does not establish either factor as the cause of the original near-uniform outputs. Exploratory forecasting results are evaluated separately. On 20 events resolving after the campaign models’ release dates, the Brier contrast between the two simulation conditions is $+0.112[+0.030, +0.192]$ under averaged replacement-evaluator readings and $+0.081[-0.003, +0.170]$ under another evaluator, with event-cluster 95% bootstrap intervals. The latter includes zero. This subset is not protected against evaluator knowledge leakage; some injections target the wrong event, and the null payloads are not length-matched. An existing single-agent baseline does not establish a simulation advantage. The contribution is an auditable case study of extraction sensitivity, with explicit boundaries on what the available data can say about forecasting and deliberation. The full report also preserves the seven-dimension design, HPC engineering, exploratory scaling, run-integrity record, complete event tables and an annotated historical figure register, separating specifications and earlier diagnostics from validated results.

Index Terms

LLM evaluation, probability extraction, multi-agent simulation, forecasting, measurement reliability, reproductibility, benchmark audit.

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I. INTRODUCTION

A social-simulation forecast involves at least two distinct transformations: agents turn an event dossier into a discussion, and an extractor turns that discussion into a probability vector. When the extractor is itself an LLM, its model, prompt, serving environment and post-processing can affect the reported forecast. A change in forecasting score can consequently reflect a change in the discussion, a change in its interpretation, or both.

ECN-BENCH was originally built to investigate whether social simulation improves forecasting relative to direct elicitation. Auditing its execution records exposed a more immediate question: how stable is the measurement used to evaluate that claim? Near-uniform probability vectors appeared frequently in the original extraction pass. Substituting the deployed extractor substantially changed those vectors on fixed evidence. The audit also uncovered a shared reference condition, failed simulations, catalogue collisions and incomplete telemetry.

This paper reports the resulting measurement study. Its primary observation is a comparison on fixed stored evidence; its secondary observations describe repeat variability and existing follow-up experiments. Forecasting contrasts are retained as exploratory measurements of the complete pipeline. They are not evidence that a swarm has acquired knowledge unavailable to its members, nor that 300-agent deliberation improves on a single model call.

The contributions are:

- 1) An artifact-verifiable comparison of two extraction configurations on 132 simulated units, with the evidence, question, options and outcome metadata checked for equality before pairing.
- 2) A reanalysis of repeated extraction, a proxy truncation/enforcement grid and a single-agent baseline, with failures and unequal measurement budgets made explicit.
- 3) A reproducible account of how execution defects, statistical dependence and retrospective knowledge exposure constrain a forecasting benchmark’s claims.

All analyses in this revision use existing artifacts. This is a retrospective case study, not a preregistered confirmatory experiment. The aim is to describe a consequential measurement failure mode without assigning an unsupported mechanism to it.

a) Reading the full report.: The main sections present current evidence and its interpretation. The technical appendices restore the complete design scope: event and condition specification, all seven rubric dimensions, inference assumptions, source-code audit, persona and run integrity, evidence construction, HPC engineering, scaling matrices and computational costs. A generated numerical appendix lists every question and all 30 stored IDs for all four campaigns, explicitly flagging the T1 mismatch. The final register preserves all 30 original figures with corrected interpretations rather than silently discarding the visual record. Historical diagnostics are clearly separated from current estimates; they are not additional validated findings.

II. RELATED WORK

LLM-as-a-judge research already treats evaluators as imperfect measurement devices. Zheng et al. [2] document biases in LLM judgments; Haldar and Hockenmaier [3] directly investigate inconsistency across repeated judgments. We do not claim that repeated-judge instability is new. Our narrower contribution concerns the extraction of categorical probabilities from social-simulation evidence and the associated source-code and data-provenance audit.

Proper scoring rules provide a framework for evaluating probabilistic forecasts. Gneiting and Raftery [1] distinguish calibration from sharpness: sharpness concerns the concentration of the predictive distribution and does not depend on which outcome is eventually observed. A higher Brier score alone cannot locate an error in either the agents’ reasoning or the extractor’s interpretation. Likewise, a sharp distribution is not necessarily a faithful or accurate one.

OASIS [4] supports large-scale LLM social simulations. ECN-BENCH uses the MiroFish-Offline pipeline built on that framework; the simulation infrastructure is the testbed for this audit, rather than a new simulation algorithm. ForecastBench [5] evaluates questions whose answers are unknown when forecasts are submitted. That prospective design addresses a limitation our retrospective release-date filter cannot resolve: every LLM component capable of generating a forecast must be considered when evaluating knowledge leakage.

A. Benchmarks, agents and social interaction

Knowledge-oriented evaluation such as MMLU [6] differs from environment-interaction benchmarks such as SWE-bench [7] and WebArena [8]. ECN-BENCH studies a further composition: multiple model-generated agents interact, after which a separate model extracts a forecast. Success of one stage does not certify another, and the benchmark does not claim to measure all forms of agent competence.

Generative-agent research [9] motivates memory and interaction architectures, while multi-agent debate [10] studies structured disagreement in reasoning tasks. Sycophancy research examines agreement with an interlocutor’s position [11]. These provide candidate mechanisms and design comparisons; they do not establish a mechanism for the present recorded forecasts. The action-count and persona audits are not validated measurements of human social behavior.

B. Memory systems and serving infrastructure

GraphRAG [12] motivates graph-indexed retrieval and summarization. The project uses Graphiti/Neo4j-based temporal memory in its execution stack, but no matched memory ablation here establishes a forecasting benefit. The archived source, rather than a contemporary software landing page, defines the relevant implementation. PagedAttention and vLLM [13] address serving efficiency; their use does not guarantee semantic output correctness or consistency across local and hosted recovery configurations.

C. Forecasting systems and contamination tests

Halawi et al. [14] study a retrieval-augmented forecasting system with forecast aggregation, rather than a universal single-call baseline. Its prospective evaluation goals are distinct from this retrospective audit. Oren et al. [15] develop a black-box test-set contamination procedure; a release-date filter or a low-Brier threshold is not an equivalent test. The report retains the broader forecasting research motivation while declining to claim leakage-free performance from the present evidence.

III. TESTBED AND EVIDENCE PROVENANCE

A. Events and conditions

The analysed catalogue contains 30 resolved events curated from Polymarket, with categorical outcome sets of size $K = 2-9$. We retain the catalogue’s identifiers: 15 C events, seven S events and eight T events. These labels are record identifiers; we do not treat their historical topical grouping as a validated taxonomy. The catalogue’s descriptive header contains superseded counts, so the executable analysis validates the observed campaign units.

Each campaign contains 30 records for each of three conditions:

- **Condition A: shared direct reference.** The stored reference evidence is reused across campaigns. It is not a separately executed baseline for each campaign model.
- **Condition B: nominally relevant injection.** A configured 300-agent, 60-round simulation receives an event-specific update at round 30.
- **Condition C: null injection.** The same configured workflow receives an unrelated administrative update at round 30.

A unit is a campaign–event–condition record. Thus 360 unit records exist, but neither the four copies of the shared reference nor repeated evaluator readings constitute independent simulation replications. Agent count is a configuration parameter, not the number of independent human-like observations or a guarantee that every agent contributed substantive discussion.

The intended B/C manipulation was relevant versus unrelated information. The released inputs implement a less controlled comparison. Seven injection identifiers are aligned to a stale catalogue. Among the 20 post-release events used later, C6 concerns a papal conclave but receives a Fed update, T6 concerns the Irish election but receives a GPT-5 update, and T8 concerns the Canadian election but receives a Bitcoin update. Repairs corrected simulation questions while retaining this injection bank. Consequently B is described throughout as *nominally* relevant.

The bank also fails the intended length match: relevant-update bodies contain 1.51–2.37 times as many characters as null bodies (median 2.04), and all null bodies use the same text. This is a measurement of the released input bank, not an estimate of the subsequent discussion length. Relevance, wording, length and stimulus diversity are therefore not separated by this design.

B. Campaign identity and recovery

The four campaign labels are Llama-3.1-8B, Mistral-7B, Qwen2.5-7B and Qwen2.5-14B. The Llama campaign originally used an AWQ-INT4 checkpoint under local vLLM; its recovery used a hosted Llama endpoint. Mistral used the v0.1 instruct model with local full-precision serving; Qwen2.5-7B used a local Ollama/GGUF deployment. The Qwen2.5-14B artifacts span original and recovery executions, and their names and serving descriptions are not consistent enough to establish one numerical-precision setting for the whole campaign. We retain the label without treating it as a controlled size or quantization comparison. Release generation, serving and precision are confounded across campaigns.

The Llama backend outage affected 23 events; catalogue correction brought the recovery set to 24 events. An initial recovery incorrectly constructed empty evaluator evidence and was superseded. The final stored recovery records contain rebuilt evidence and replacement scoring. Only the six unaffected events (C3, S2–S6) can enter the comparison with the original July extractor.

The full question-level audit additionally finds Qwen2.5-14B T1 still asks about a Fed decision rather than the UAW question carried by the other campaigns. Its options and outcome happen to match, so vector validation does not detect the topic collision. T1 is outside the primary 20-event subset; it is excluded across all campaigns in the extended 29-event question-aligned summaries, while its stored rows remain visible.

Qwen2.5-14B required six wrong-topic events to be re-simulated before the present revision. A separate unit, C9 in Condition C, retains an outage-like telemetry signature. Its scored probabilities survive, but its simulation integrity is uncertain. We report full-set descriptive results and a sensitivity analysis excluding the entire C9 event across campaigns, alongside the wrong-injection events, to keep a balanced comparison.

C. Extractor identities and read budgets

The archive separates evaluator files:

- **Evaluator J:** the original local DeepSeek-R1-Distill-Qwen-14B deployment, identified in the pipeline as `deepseek-r1-14b`.
- **Evaluator A:** the recorded hosted identifier `deepseek/deepseek-v4-flash-0731`. Three campaigns have an initial pass and three additional reads. Qwen2.5-14B has three reads only.
- **Evaluator C:** the recorded hosted identifier `openai/gpt-5.6-luna-pro`, with one read per unit.

These hosted names are execution identifiers from the archive, not evidence that the same backend or weights remain available. Provider routing and backend determinism were not independently verified.

The file `event_results.json` is not uniformly Evaluator J data: Llama recovery rows and 18 Qwen2.5-14B recovery rows use a replacement `deepseek/deepseek-r1` evaluator. The fixed-evidence J/A comparison excludes these substituted readings and excludes Qwen2.5-14B entirely.

Four known failed rereads survive as stale copies of previous values: Qwen2.5-7B’s T3/C, C9/A and C13/C in repeat 3, and Qwen2.5-14B’s T1/C in repeat 2. The analysis excludes exactly these file–unit pairs. All 2,070 stored campaign probability vectors pass finite-value, range, normalization and option/outcome consistency checks. This validates record structure; it does not certify execution integrity or independently verify the historical event outcomes.

IV. MEASUREMENT AND ANALYSIS

A. Estimand and scoring

Let T_{mec} denote the stored evidence for campaign m , event e and condition c . An extraction configuration j produces $\hat{\mathbf{p}}_{mecj} = g_j(T_{mec}, q_e, \mathcal{O}_e)$, where q_e is the question and $\mathcal{O}_e$ the outcome list. The extraction function includes the model, prompt, generation settings, retries, serving and post-processing. The final forecast is an output of this composite system.

We use the unnormalised multicategory Brier score:

$$BS(\mathbf{p}, y) = \sum_{o \in \mathcal{O}} (p_o - \mathbf{1}\{o = y\})^2, \quad 0 \leq BS \leq 2. \quad (1)$$

For a uniform vector this equals $1 - 1/K$. A uniform vector is a legitimate probabilistic forecast; it is not automatically an error or a missing answer. The term *near-uniform* is an operational output diagnostic:

$$F(\mathbf{p}) = \mathbf{1} \left\{ \max_o |p_o - 1/K| < 0.005 \right\}. \quad (2)$$

We also report sensitivity to this tolerance. Near-uniform units remain in all forecasting score summaries. Excluding them because they are uninformative would change the estimand and could selectively remove difficult cases.

Directional accuracy credits a tie among h maximal-probability options by $1/h$ if the outcome belongs to that tie, and zero otherwise. The mean $1/K$ is reported as a descriptive uniform-choice reference, not a competitive forecasting baseline and not a null model for substantive event predictability.

B. Repeated readings and aggregation

For Evaluator A, the principal descriptive forecast uses the per-unit mean of all available probability vectors. Brier is then computed on that mean. A sensitivity analysis instead averages the Brier scores of individual reads. These are different quantities:

$$\frac{1}{J} \sum_j BS(\mathbf{p}_j, y) = BS(\bar{\mathbf{p}}, y) + \frac{1}{J} \sum_j \|\mathbf{p}_j - \bar{\mathbf{p}}\|^2. \quad (3)$$

Averaging vectors therefore changes the forecast and can improve its score through ensembling. It does not simply reveal an otherwise hidden true swarm forecast. Evaluator C has only one read, so comparisons against the Evaluator A ensemble also differ in measurement budget.

Within-unit Brier range and exact vector agreement describe repeatability. Exact agreement is a strict diagnostic; disagreement can be numerically small, and range tends to increase with the number of reads. These quantities should not be used as calibrated thresholds for which population effects are measurable. Nor does inter-evaluator correlation establish agreement or validity.

C. Dependence and uncertainty

The four campaigns answer the same events. We first form each event's mean paired contrast across the four fixed campaigns, then resample events with replacement. All observations belonging to a sampled event are kept together. We use 100,000 percentile bootstrap replicates, NumPy's generator with seed 7, and 2.5th/97.5th percentiles. The estimand is the equal-event average over these four campaigns, not an average over a population of independently sampled models. Intervals condition on the observed simulations and evaluator readings.

The 20-event subset retains the prior analysis's event selection to permit comparison. The selection predates this revision but is not preregistered. We call it the *post-release subset*, never contamination-clean: its outcomes resolve after the stated campaign-model release dates, but no corresponding protection exists for the later evaluator deployments or for retrospectively authored inputs.

Event bootstrap still assumes independent event clusters. C1, C5 and S2–S5 share the 2024 US election. We therefore additionally group these six events into one cluster (15 clusters for 20 events), and separately omit them (14 events). Grouped resampling preserves all events in a selected cluster and computes their event-weighted mean. These sensitivity choices are retrospective and do not exhaust every possible dependency in the catalogue.

We do not report the previous pooled Poisson-binomial p -value. It treated repeated model–event records as independent and rounded a fractional tie-credit total to an integer threshold. Neither operation supplies the exact sampling distribution of the reported accuracy statistic. More generally, no analysis in this paper is offered as a multiplicity-adjusted confirmatory test. Intervals across models, evaluators, variants and exclusions describe sensitivity rather than a search for one significant result.

V. FIXED-EVIDENCE EXTRACTION SENSITIVITY

Table I reports the primary audit. Mistral and Qwen2.5-7B each contribute all 60 simulated B/C units. Llama contributes 12 units from its six unaffected events. The analysis verifies equality of stored evidence text, question, option list and ground truth for each pair, and aborts on mismatch.

TABLE I
NEAR-UNIFORM READINGS ON MATCHED SIMULATED EVIDENCE. COUNTS REFER TO THE SAME UNITS WITHIN EACH ROW.

Campaign	Units	Evaluator J	Evaluator A
Llama-3.1-8B	12	12	0
Mistral-7B	60	19	1
Qwen2.5-7B	60	0	0
Total	132	31	1

The change from 31/132 to 1/132 establishes that the reported probability vectors are not invariant to extraction configuration. In the unaffected Llama subset all 12 original readings are near-uniform and none of their replacements are. This subset was selected by execution integrity, not randomly; it establishes that a large difference can occur, not its prevalence in arbitrary simulations.

The counts are unchanged using numerical exact-uniformity (maximum deviation at most 10^{-12}) or tolerance 0.01; at tolerance 0.02 the J count becomes 32 while the A count stays one. The contrast is therefore not generated by the particular 0.005 threshold in this paired material.

This is a fixed-*evidence* comparison, not a byte-identical HTTP-request experiment. Model, serving, generation budget, schema handling and possibly prompt/validation details differ across the historical configurations. The event-specific micro-question catalogue also underwent correction. The comparison locates sensitivity in the extraction stage as deployed; it does not isolate evaluator architecture from those other factors.

Most importantly, a non-uniform replacement is not automatically a more faithful reading. No human-labelled reference distribution or known-consensus synthetic transcript set was used to decide which extractor represents the discussion correctly. The replacement may recover information omitted by the first extractor, import outside knowledge, or become overconfident. The present records cannot distinguish these explanations.

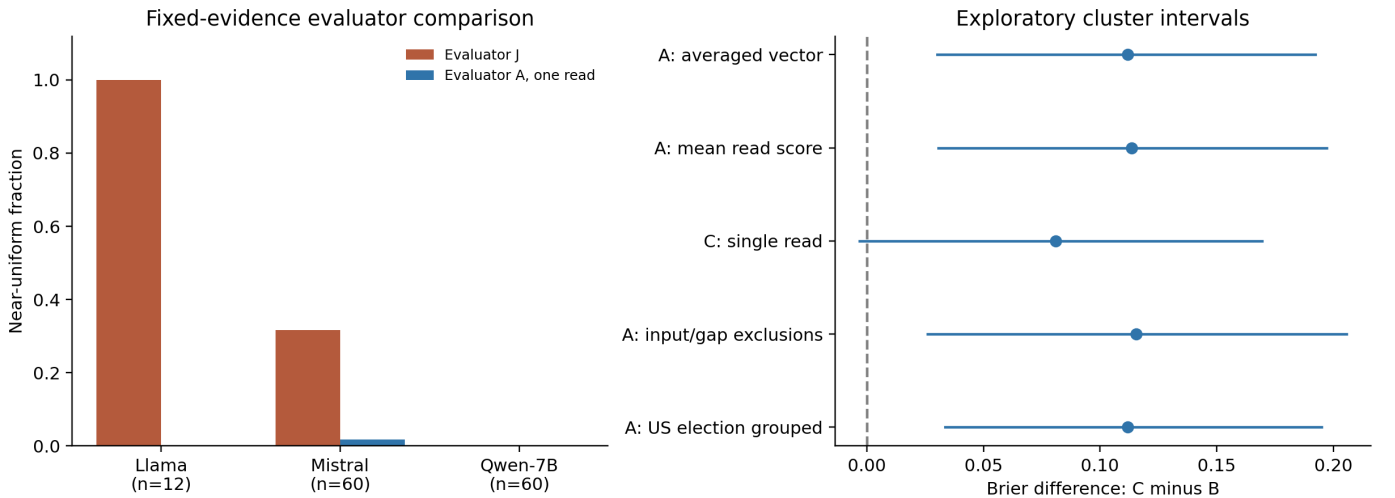


Fig. 1. Left: fixed-evidence near-uniform counts expressed as fractions (Table I). Right: exploratory Brier contrasts under alternative extraction and clustering choices (Table V). Bars on the left are observed fractions, not independent-agent estimates. Intervals on the right account for repeated campaigns within events; the grouped row additionally joins six US-election events.

A. Repeatability within the replacement evaluator

Table II shows substantial variability in repeated extraction. Mean within-unit Brier ranges are 0.208–0.375. Repeated reads here are repeated requests through the recorded deployment, not independent evaluators and

TABLE II
EVALUATOR A REPEATABILITY ACROSS 90 UNITS PER CAMPAIGN. READS ARE AVAILABLE FILES; FOUR STALE FILE-UNIT READINGS ARE EXCLUDED. EXACT AGREEMENT CONCERNS PROBABILITY VECTORS, NOT ROUNDED BRIER VALUES.

Campaign	Reads	Exact	Mean BS range
Llama-3.1-8B	4	3/90	0.330
Mistral-7B	4	0/90	0.375
Qwen2.5-7B	4	4/90	0.280
Qwen2.5-14B	3	3/90	0.208

not new simulations. The archive does not establish whether stochastic decoding, retries, backend nondeterminism or provider routing generated each difference. Because some requests failed and their stale copies are excluded, read counts are not perfectly balanced even within a campaign. The ranges support reporting repeatability alongside forecast quality; they do not provide a universal noise floor or an optimal number of required reads.

VI. SOURCE AUDIT AND EXISTING MECHANISM PROBES

A. What the code establishes

The archived client includes a keyword-based bypass for model identifiers containing strings such as `r1` or `reasoning`, and a separate local-quantization bypass. The identifier `deepseek-r1-14b` matches the reasoning filter. This documents a path that disables constraints for the configured name. Complete historical request logs are unavailable, so actual constraint status cannot be independently verified for every July request.

Schema enforcement and semantic correctness are different properties. JSON-object mode guarantees neither the nested schema nor meaningful probabilities. A schema-valid vector can be uniform, incorrect or unrelated to the transcript. Evaluator A used JSON-object handling; Evaluator C’s strict-schema requests were documented as falling back after schema rejection. Neither is a verified, uniformly strict schema-enforced reference instrument.

Silent defaults complicate diagnosis. Historical validation could accept incomplete micro-question mappings and fill them with missing-value labels. A deployed normalization path could also replace non-positive matched mass with a uniform vector. The final probabilities do not consistently preserve the raw completion and retry history needed to assign the origin of every uniform vector. The observations cannot establish that uniformity was a deliberate hedge by the model.

B. Cross-evaluator transfer

An existing follow-up sent Mistral’s 60 simulated evidence records to five additional unconstrained deployments. The stored experiment reports 8 near-uniform outputs among 58 parsed Qwen3-14B responses; six of these eight occur among the 19 units that Evaluator J flattened. Other deployments report zero near-uniform outputs among 56 DeepSeek-R1, 54 DeepSeek-R1-Distill-Llama-70B, 57 Qwen3-32B and 60 Ministral-14B responses.

These are counts among usable responses, not success rates over all attempted calls. Missing responses cannot be assumed missing at random. The simplified transfer prompt differs from the production rubric; an additional DeepSeek-R1 production-prompt control reports zero near-uniform outputs among 60 scored units. This control does not make every cross-model comparison configuration-matched. The overlap identifies inputs on which two deployments return the same diagnostic pattern. It does not establish a model-size boundary: size, training and serving are not independently varied.

C. Truncation and requested enforcement

The existing grid varies four evidence lengths and whether output constraints are requested for a hosted Qwen3-14B proxy. Truncation retains parts of the log and subsamples representative actions; it changes content and speaker coverage together with character count. The original Evaluator J deployment was not available for this experiment.

TABLE III

QWEN3-14B GRID: NEAR-UNIFORM/USABLE READINGS OUT OF 60 ATTEMPTED UNITS PER CELL. ON DENOTES REQUESTED ENFORCEMENT; ACTUAL STRICT-SCHEMA ACCEPTANCE IS NOT VERIFIED FOR EVERY RESPONSE.

Length target	Off	On
2k	7/59	5/59
5k	4/60	2/60
8k	0/60	4/60
full	1/60	0/60

The unconstrained shortest arm has 7/59 near-uniform readings and the full arm 1/60. Pairing the 59 units with both responses yields six short-only near-uniform outcomes and zero full-only outcomes. Shortening did not reduce near-uniformity in this proxy experiment. A descriptive association between longer evidence and near-uniformity in the original Mistral/J observations cannot be promoted to a demonstrated length mechanism.

Across all lengths there are 238 matched off/on pairs because missing units differ between arms. We average paired indicator differences within each event and resample the 30 events, preserving conditions and length levels. The equal-event off-minus-on estimate is approximately $+0.005$, with interval $[-0.029, +0.042]$. The marginal totals, 12/239 and 11/239, should not be treated as independent binomial samples. This experiment does not detect a clear effect of requesting enforcement; it neither proves equivalence nor establishes that enforcement has no effect in the original deployment.

These probes bound candidate explanations without identifying the cause of Evaluator J’s outputs. The extraction-sensitivity observation does not require either a length mechanism or a schema mechanism to be true.

VII. EXPLORATORY FORECASTING RESULTS

A. Post-release subset

The 20 selected events yield 80 campaign–event records per simulation condition. Under Evaluator A’s averaged vector, B accuracy is 81.9%, comprising 65 outright correct modal outcomes and one two-way tie credited by one half. The mean uniform-choice reference is 32.6%. These are descriptive statistics of retrospective pipeline outputs.

TABLE IV

POST-RELEASE SUBSET, 20 EVENTS PER CAMPAIGN. ACCURACY USES TIE CREDIT; BRIER USES EVALUATOR A’S AVERAGED VECTOR. LABELS DO NOT IMPLY MATCHED SERVING.

Campaign	Acc. B	Acc. C	BS B	BS C
Llama-3.1-8B	0.675	0.500	0.545	0.613
Mistral-7B	0.850	0.650	0.378	0.441
Qwen2.5-7B	0.850	0.550	0.348	0.520
Qwen2.5-14B	0.900	0.800	0.289	0.432

The contrast $BS_C - BS_B$ is positive when B scores better. Table V presents all sensitivity summaries selected for this retrospective reanalysis; none is a preregistered hypothesis test.

The averaged-vector and mean-read-score summaries under A are similar here, so the B/C estimate is not solely a vector-ensembling artifact. Excluding the input/gap events retains a positive estimate. Grouping or omitting six US-election events also retains a positive estimate under A. These checks do not remove every shared cause of event outcomes.

Evaluator C gives $+0.081[-0.003, +0.170]$, whose interval includes zero. Its single-read budget differs from A’s repeated-read budget. This is limited robustness across the available measurement choices, not a clean test that one evaluator detects an effect and another does not. A statistical difference between the two effects is not established.

TABLE V

EXPLORATORY SENSITIVITY OF $BS_C - BS_B$. INTERVALS RESAMPLE EVENTS, EXCEPT THE US-ELECTION-GROUPED ROW (15 CLUSTERS OVER 20 EVENTS). INPUT/GAP EXCLUSION REMOVES C6, T6, T8 AND C9 FROM ALL CAMPAIGNS. EXCLUSIONS WERE CHOSEN AFTER INSPECTION OF THE AUDIT TRAIL; INTERVALS DO NOT CORRECT FOR THIS SELECTION.

Analysis	Events	Mean [95% interval]
Evaluator A, averaged vector	20	+0.112[+0.030, +0.192]
Evaluator A, mean read score	20	+0.114[+0.031, +0.197]
Evaluator C, single read	20	+0.081[−0.003, +0.170]
A, excluding input/gap events	16	+0.116[+0.026, +0.206]
A, US election grouped	20	+0.112[+0.034, +0.195]
A, excluding US election	14	+0.117[+0.013, +0.218]

B. What the contrast cannot identify

The post-release filter concerns the campaign models, not all LLMs in the pipeline. Evaluators deployed in 2026 may know the outcomes of events from 2024–2025. The probability prompt exposes identifiable questions, and instructions to use only the discussion do not demonstrate compliance. Dossiers and perturbations were assembled retrospectively; timestamps, source fidelity and freedom from hindsight were not independently established for every claim. Stripping URLs or softening deterministic language cannot remove knowledge already encoded in a model.

The accuracy figures therefore do not establish leakage-free forecasting. They may reflect genuine evidence use, pre-existing model knowledge, evaluator knowledge, retrospective input construction, or a combination. Ground-truth labels are used as recorded; this study does not independently adjudicate every market resolution or the mutual exclusivity of every curated option set.

The B/C contrast describes sensitivity of the complete simulated-and-extracted pipeline to its input conditions. It cannot attribute that sensitivity specifically to agent-to-agent deliberation. A direct call, an extractor reacting to a changed transcript, or different injected wording may also produce a B/C difference. Relevance is not isolated from length, and C6/T6/T8 do not implement the intended relevance manipulation.

VIII. EXISTING SINGLE-AGENT BASELINE

The post-campaign baseline contains 240 calls: 20 events, four hosted model arms and three calls per event/arm. Each receives the seed dossier and nominal B update, not the simulation transcript. The first call uses temperature zero, the other two temperature 0.7; these are not identically distributed replicates.

The principal summary averages individual-call Brier scores. It estimates the mean score of the observed three-setting mixture, not a fixed-temperature single-call policy. The script also reports the first call alone and the score of the three-vector ensemble. No added retrieval step was executed: the dossier and injection were supplied directly in the prompt.

Only Llama-3.1-8B and Qwen2.5-7B have nominal model identities matching their campaigns; identical served weights are not independently verified, and serving and precision are not fully matched. Ministral-8B and Qwen3-14B differ from the Mistral-7B and Qwen2.5-14B campaigns and are secondary comparisons.

TABLE VI

EXISTING CoT BASELINE. POSITIVE $BS_{CoT} - BS_{swarm}$ FAVOURS THE SWARM. IDENTITY MATCHING DOES NOT IMPLY SERVING/PRECISION MATCHING. PER-ARM INTERVALS ARE EXPLORATORY EVENT BOOTSTRAPS.

CoT arm	Identity matched?	Accuracy	Brier difference [95% interval]
Llama-3.1-8B	Yes	0.778	−0.181[−0.366, −0.020]
Qwen2.5-7B	Yes	0.790	−0.030[−0.134, +0.082]
Mistral-8B	No	0.912	−0.170[−0.371, +0.010]
Qwen3-14B	No	0.858	−0.008[−0.114, +0.092]

Pooling the identity-matched arms by event gives $BS_{CoT} - BS_{swarm} = -0.106[-0.226, +0.008]$. The point estimate favours CoT and the interval includes zero. This is no demonstrated swarm advantage in the available

comparison, not proof of equivalence or non-inferiority. Some individual or ensemble variants exclude zero in CoT’s favour; these are exploratory and do not change that scope.

Across all four, partly unmatched arms, individual-call accuracy is 83.4%, compared with 81.9% for the swarm under A’s averaged vectors. This does not establish equal capability: model identity, serving and aggregation are unbalanced. There is no single-agent C arm. A factorial comparison of single-agent versus simulation crossed with the same B/C inputs would be needed to estimate additional content sensitivity from deliberation. The present data do not identify that interaction.

IX. IMPLICATIONS AND LIMITS

A. *A practical measurement protocol*

A reproducible evaluation should preserve the event question, ordered options, evidence bytes, full prompt, model identifier, serving revision when available, generation parameters, requested and accepted output format, raw completion, retry history and probability post-processing. A final vector and an error flag are insufficient to diagnose the defects found here.

Repeated extraction and a second evaluator are useful diagnostics, not substitutes for external validity. Read counts should follow a precision/cost analysis; this study does not justify a universal minimum of three reads. A labelled synthetic transcript or blinded human reference can test extraction fidelity. A question-only baseline can measure what a later evaluator knows without the simulation. These are recommendations, not experiments executed in this revision. Repairing a schema bypass is sound engineering even when its effect on near-uniformity is unknown. Valid JSON, a non-uniform forecast and a faithful interpretation are separate requirements.

B. *Limits of the evidence*

The primary comparison is conditional on a selected artifact set and a historical deployment. The original endpoint’s disappearance prevents exact intervention on that deployment. An open-weight model identifier exists, but restoring weights under a new stack would be a new experiment, not a proven replay.

Repeated reads condition on one realised simulation per condition and cannot estimate simulation-run variability. Nominally 300 agents sharing a model do not supply 300 independent replications. The 30-event convenience catalogue supports limited generalization to other events, languages, forecasting horizons and deployments.

The seven-dimension rubric, convergence telemetry and persona/action audits remain execution diagnostics, not validated measures of psychological or collective-reasoning constructs. Incomplete telemetry, placeholders and serving failures prevent the original convergence and optimal-configuration claims from being retained as results. Isolated failures of small deployments do not establish a general parameter-count exclusion rule.

No causal mechanism is rescued by an insignificant comparison, and an inability to reject a null is not proof of equality. The main finding is sensitivity of measured probabilities to extraction configuration; forecasting comparisons make its consequences and limits visible.

X. CONCLUSION

On 132 matched simulated evidence records, the original and replacement extraction configurations produce 31 and one near-uniform forecasts. Repeated reads of the replacement also vary substantially. These observations warrant treating an LLM extractor as part of the forecasting system.

The source audit and proxy experiments do not identify why the original configuration produced these vectors or which reading is faithful to the swarm. Forecasting contrasts remain retrospective, evaluator-dependent and affected by imperfect inputs. An existing single-agent baseline does not demonstrate a deliberation benefit. The contribution is a reproducible audit of extraction sensitivity and the design boundaries it exposes, rather than a validated advantage or failure of collective reasoning.

APPENDIX A

EXPERIMENTAL AND BENCHMARK DESIGN

A. Taxonomy and Event Selection

ECN-BENCH curates 30 resolved categorical questions from Polymarket. Market prices are not a competitive baseline in this experiment. Recorded outcomes are used as supplied, not independently re-adjudicated. The analysed set is defined by the 30 IDs actually present in each campaign, not the superseded counts in the catalogue header.

The initial specification used information type and forecast horizon as design axes. Some catalogue entries contain such metadata. The historical analyses instead used the prefix partition below, which mixes subject matter and question format; it is retained for traceability, not as a validated taxonomy. The extended numerical appendix gives the actual questions, recorded outcomes and all per-event scores.

TABLE VII
ECN-BENCH EVENT TAXONOMY AS REALISED IN THE 30-EVENT CORPUS

Prefix	Category	Events	Option count K
C	Social/Electoral	15	3–9
S	Senate/Binary	7	2–4
T	Technical/Economic	8	2–5
Total		30	2–9

B. Experimental Conditions

The implemented conditions are listed below. They do not isolate a causal effect of deliberation because the direct reference is shared and the B/C controls are imperfect:

- **Condition A (shared direct reference):** Seed-based reference evidence is reused across campaigns. This is not an independently executed baseline for each campaign model.
- **Condition B (nominally relevant injection):** A configured 300-agent, 60-round simulation receives an event-specific update at round 30. Seven injection IDs target a stale catalogue; relevant is therefore the intended rather than universally realised manipulation.
- **Condition C (null injection):** The same configured workflow receives an unrelated administrative message at round 30. Null payloads repeat the same body and are not length-matched to the B payloads.

C. Simulation Output Fidelity: Action Density and Scope Conditions

The original project reported a cluster-side scan of 1,380 SQLite database files. The two tables preserve that historical scan, including sweeps and wrong-topic Qwen2.5-14B executions; they are not a matched comparison of the four currently scored campaigns. The complete original collection is no longer available for rechecking. A cursor-tracking issue in actions summaries can omit organic actions during telemetry rounds without removing them from the SQLite trace table.

Action counts do not establish deliberation quality, antagonism, human-like cognition or causal forecasting mechanisms. Unequal execution collections cannot rank campaigns. The original all-campaign scan is distinct from the run-integrity audit and the currently scored evidence; in particular, its Qwen2.5-14B row includes executions against subsequently corrected questions.

APPENDIX B

THE SEVEN-DIMENSION GRADING FRAMEWORK

This section restores the complete seven-dimension design specification. It is not a psychometrically validated scale or seven successfully measured outcomes. Probability-vector Brier scores are recomputed; the remaining dimensions require observables and construct validation that the campaign does not consistently supply. Original weights and display bands document the intended protocol, not empirically justified weights.

TABLE VIII

HISTORICAL CLUSTER-SCAN SOCIAL-ACTION COUNTS FROM SQLITE TRACE TABLES (1,380 DATABASE FILES, MIXED CAMPAIGNS AND SWEEPS). THE `ACTIONS.JSONL` SUMMARY FILES UNDERCOUNT ORGANIC ACTIONS DUE TO A CURSOR-TRACKING BUG DURING TELEMETRY-PROBE ROUNDS (SEE TEXT); THE `.DB` FILES ARE THE AUTHORITATIVE SOURCE.

Model	DBs scanned	Total Social Actions	Avg/DB	Agent Traces
Llama-3.1-8B-AWQ	672	27,087	40.31	554,782
Mistral-7B-Instruct-v0.1	264	9,364	35.47	257,339
Qwen2.5-14B	180	2,407	13.38	113,366
Qwen2.5-7B-GGUF	264	723	2.74	191,942

“Social Actions” = post + comment + like + comment_like + follow + dislike + comment_dislike (excluding telemetry probes and refresh/sign_up). “Agent Traces” = all trace-table rows including telemetry. Llama-3.1-8B’s 672 DBs include the main

July campaign (180) plus optimization sweeps and earlier runs; per-DB averaging normalizes across unequal DB counts.

Qwen2.5-14B’s 180 DBs = a reported 30-event execution collection (30 events $\times$ 3 conditions $\times$ 2 platforms).

Reproducibility caveat: these counts were obtained by SQL queries executed on the cluster and are not recoverable from the released `event_results.json` artifacts, which do not carry action counts. They could not be re-verified during the audit described in Appendix J and are reported on the authority of the original scan. Exporting per-unit action counts into the released summaries is a required change for the next campaign. Note also that the Llama-3.1-8B row pools heterogeneous configurations (different N and R across sweeps), so its per-DB average is not directly comparable with rows drawn from a single fixed configuration.

TABLE IX

SOCIAL-ACTION BREAKDOWN BY TYPE (HISTORICAL `.DB` SCAN, MIXED CAMPAIGNS AND SWEEPS).

Action	Llama-8B	Mistral-7B	Qwen-14B	Qwen-7B
post	8,079	4,476	1,215	592
comment	5,421	1,978	585	108
like	4,567	1,946	461	16
comment_like	3,010	167	124	1
follow	5,993	708	11	1
dislike	7	67	0	5
comment_dislike	10	22	11	0
Total	27,087	9,364	2,407	723

A. D1: Prediction Accuracy (Weight: 0.25)

The primary forecasting metric is the **Multi-Category Brier Score (BS)**, which evaluates the extracted probabilistic forecast. For a given event with K mutually exclusive categories, let f_k be the swarm’s predicted probability for category k , and o_k be the one-hot indicator of the realized outcome ($o_k = 1$ if category k occurred, 0 otherwise). The Brier Score is formulated as:

$$BS = \sum_{k=1}^K (f_k - o_k)^2 \quad (4)$$

The unnormalised multicategory Brier score lies in $[0, 2]$. A correct deterministic forecast attains zero; a deterministic forecast on one wrong option attains two. Uniform probability scores $1 - 1/K$ for every outcome. Brier measures overall probabilistic performance, not calibration alone. All current scores are recomputed from vectors rather than inherited scalar fields.

The aggregate Brier Score across E events is $BS_c = \frac{1}{E} \sum_{e=1}^E BS_{e,c}$ for condition $c \in \{A, B, C\}$.

We formally define **Simulation Lift (Lift)** as the historical difference in Brier score:

$$Lift = BS_A - BS_B \quad (5)$$

A positive value means B scores lower than the shared A reference. It is not a within-model simulation effect, even if an interval excludes zero. Evaluator identity and input provenance remain part of the comparison. The principal observable input contrast is $BS_C - BS_B$, with its remaining confounds stated explicitly.

The specification also proposes Brier skill relative to a uniform vector. For pooled reporting, averaging eventwise ratios differs from dividing mean Brier by mean reference; the aggregation must be specified. The main paper uses unnormalised Brier with equal event weights rather than claiming normalization removes all outcome-space effects:

$$BSS = 1 - \frac{BS_{model}}{BS_{random}} \quad \text{where} \quad BS_{random} = 1 - \frac{1}{K} \quad (6)$$

Where BS_{random} represents the expected multi-category Brier error under uniform random guessing ($f_k = \frac{1}{K}$). The BSS metric possesses intuitive econometric bounds:

- $BSS = +1.0$: Perfect deterministic prediction ($f_{correct} = 1.0$).
- $BSS = 0.0$: Performance equivalent to uniform random guessing.
- $BSS < 0.0$: Sub-random performance (worse than uninformed guessing).

The specified display score is $D1_{score} = \max(0, 100 BSS_B)$. Clipping hides differences below the reference and is not itself a proper scoring rule. It is retained as an interface convention, not used for inference or model selection.

B. D2: Polarization (Weight: 0.15)

The construction below is inspired by Esteban and Ray [16]; its vector-valued extension and clustering choices are specific to the benchmark proposal. To measure whether the agent swarm fragments into deeply entrenched, opposing camps, we utilize the **Esteban-Ray (ER) Index of Polarization**. Rather than measuring general dispersion, the ER Index captures both identification (internal group cohesiveness) and alienation (distance between groups). Assume the simulated population forms G distinct opinion clusters regarding the event outcome. Let p_i be the population proportion of cluster i , and y_i be the mean probability vector of group i . The ER Index is defined as:

$$ER(\alpha) = \kappa \sum_{i=1}^G \sum_{j=1}^G p_i^{1+\alpha} p_j \|y_i - y_j\|_1 \quad (7)$$

Where:

- α is the polarization sensitivity parameter. The protocol selects $\alpha = 1.0$; this is not an empirically calibrated psychological parameter.
- κ is a normalization constant scaling $ER \in [0, 1]$.
- The clustering procedure and normalization must be specified before computing this index.

This functional increases with separation under its chosen clustering, distance and normalization. Its vector-valued extension and nominal score $S_2 = 100(1 - ER_{norm})$ are protocol choices. A reproducible implementation requires an explicit clustering rule and normalization; a generated high/low label is not a measured real-population polarization index.

C. D3: Herd Effect (Weight: 0.15)

The proposed herd diagnostic counts changes in modal answers toward the majority. This is a behavioral transition statistic, not a causal measure of social pressure: shared evidence, common model priors and a moving majority can generate the same pattern. The original indicator is:

$$Conform_{i,t} = \mathbf{1} [\arg\max_k p_{i,t,k} = M_t \quad \text{and} \quad \arg\max_k p_{i,t-1,k} \neq M_{t-1}] \quad (8)$$

The raw herd index for a simulation is:

$$Herd = \frac{1}{T \cdot N_{agents}} \sum_{t=1}^T \sum_{i=1}^{N_{agents}} Conform_{i,t} \quad (9)$$

The proposed B/C contrast is:

$$D3_{raw} = Herd_B - Herd_C \quad (10)$$

Neither this sign nor a raw count separates mimicry from rational updating. A strong response to relevant information may be appropriate. Computing the indicator requires longitudinal agent probabilities and a tie convention, not a substitute evaluator impression.

D. D4: Deliberation Quality (Weight: 0.15)

The conceptual source is the discourse-quality framework of Steenbergen et al. [18]; the shortened LLM rubric is an adaptation, not a validated reproduction of that instrument. The design adapts discourse-quality ideas into the criteria below. It originally targeted a stratified sample of 100 interactions; the surviving evidence uses a capped, ranked sample instead. This is not proof that 100 random interactions were evaluated for every unit:

- 1) **Participation:** Equality and accessibility of speech acts.
- 2) **Level of Justification:** Whether claims are accompanied by logical assertions (0: no justification; 1: inferior; 2: qualified; 3: sophisticated).
- 3) **Respect:** Tone toward peers (0: disrespectful; 1: neutral; 2: explicitly respectful).
- 4) **Constructive Politics:** Collaboration toward a consensus (0: positional; 1: alternative proposals; 2: mediating consensus).

The normalized score is $S_4 = 100 \times \frac{Score_{observed} - Score_{min}}{Score_{max} - Score_{min}}$.

E. D5: Susceptibility (Weight: 0.10)

The proposed susceptibility measure contrasts pre/post Brier changes in B and C. It requires valid pre-injection probabilities on the same outcome space and matched measurement at both times. These formulas specify the intended quantity, not an estimand established by final-output records alone:

$$PostImprove_{e,m,c} = BS_{e,m,c,(t=29)} - BS_{e,m,c,(t=60)} \quad (11)$$

Where $t = 29$ is the step immediately preceding the perturbation. The raw Susceptibility score is the difference in lift:

$$D5_{raw} = \frac{1}{E} \sum_{e=1}^E (PostImprove_{e,m,B} - PostImprove_{e,m,C}) \quad (12)$$

A positive value describes a larger measured score improvement in B conditional on valid measurements. It does not mathematically establish rationality or noise resistance. The current analysis uses the observable final-output contrast and does not invent round-29 measurements.

F. D6: Convergence (Weight: 0.10)

DeGroot’s linear opinion-pooling model [17] motivates the notation; the LLM simulation has not been shown to obey that model. We model consensus formation through the **DeGroot Consensus Model**. In the classical model, agent beliefs evolve via a transition probability matrix W representing social trust, such that $x(t+1) = Wx(t)$. In ECN-BENCH, we measure the mathematical distance to consensus by the final round. First, we define the dispersion trajectory of agent probability vectors at step t :

$$Disp_t = \frac{1}{N_{agents}(N_{agents} - 1)} \sum_{i < j} \sum_{k=1}^K |p_{i,t,k} - p_{j,t,k}| \quad (13)$$

The opinion convergence (contraction rate) is defined as:

$$Contr = 1 - \frac{Disp_T}{Disp_1} \quad (14)$$

To prevent the swarm from converging on the wrong answer (i.e., highly confident but incorrect consensus), we implement an **Accuracy Guard**:

$$Guard = 1 - BS_B \quad (15)$$

The convergence score is formulated as the joint product:

$$D6_{raw} = \text{mean}_e(Contr_e \times Guard_e) \quad (16)$$

Under the adopted convention, $1 - BS_B$ ranges from -1 to 1 , not 0 to 1 . The contraction ratio is undefined at zero initial dispersion and may be negative. The original dispersion sum is over $i < j$ but divided by $N(N-1)$, yielding half the mean unordered-pair distance; consistent scaling cancels in its ratio. Incomplete telemetry does not validate this composite.

G. D7: Information Diversity (Weight: 0.10)

The diversity component proposes entropy over semantic argument bins. Its interpretation depends on an explicit binning and coding procedure, not the number of agents. The released categorical judgments do not establish that a reproducible semantic-binning procedure was executed:

$$H = - \sum_{i=1}^S q_i \ln(q_i) \quad (17)$$

To normalize this across varying bin sizes, we compute Pielou’s Evenness (Shannon Equitability):

$$E_H = \frac{H}{\ln(S)} \quad (18)$$

The intended aggregate is $S_7 = 100 \text{ mean}_{e,t} E_{H,e,t}$. Zero-frequency terms contribute zero. A single bin requires an explicit convention because $\ln S = 0$, and an empty sample is missing rather than maximally diverse. Diversity alone is not accuracy or independent information.

H. Composite Aggregation and Grade Bands

The final **ECN-Predictability Index (EPI)** is a weighted composite of the seven normalized dimension scores ($S_1 \dots S_7 \in [0, 100]$):

$$EPI = 0.25S_1 + 0.15S_2 + 0.15S_3 + 0.15S_4 + 0.10S_5 + 0.10S_6 + 0.10S_7 \quad (19)$$

The resulting *EPI* was intended to map to the project-defined display bands, as defined in Table X.

TABLE X
ORIGINAL PROPOSED DISPLAY BANDS (NOT VALIDATED ACADEMIC GRADES)

EPI Range	Grade	Interpretation
90 - 100	A+	Exceptional multi-agent forecasting and dynamic deliberation
80 - 89	A	Strong, well-calibrated, and resilient swarm intelligence
70 - 79	B	Solid performance, with minor miscalibrations or conformity issues
60 - 69	C	Mixed/unstable quality; susceptible to noise
50 - 59	D	Weak simulation lift; poor deliberation quality
< 50	F	Fails baseline forecasting expectations

The EPI requires seven valid normalized inputs. Missing or unstable dimensions cannot be silently filled with zeros or uniform buckets, nor dropped while retaining the original interpretation. No EPI ranking is reported. Rubric repeatability is a diagnostic, not construct validation.

APPENDIX C

FULL PROTOCOL, ESTIMANDS AND MEASUREMENT BOUNDARIES

A. Original research questions and their operational limits

ECN-BENCH was designed around three questions: whether simulation improves a direct forecast, whether agent beliefs converge, and whether relevant information changes beliefs differently from a null update. These questions remain part of the research programme. The audit distinguishes them from what the released measurements identify.

The direct-forecast question requires an independently executed baseline for each campaign model, with comparable inputs, serving and evaluation. A shared reference does not implement that comparison. The convergence question requires valid longitudinal probabilities and a defined distance between distributions; a generated rubric label or a constant initialized telemetry series does not implement it. The information-response question requires interpretable B/C inputs. The recorded injection bank changes length and diversity as well as intended relevance, and includes wrong-topic inputs. None of these conditions can be repaired retrospectively by changing a statistical test.

The configured 300 agents and 60 rounds are properties of the execution design. They are not 300 independent forecast replications and do not enlarge the event sample. Three hundred model-generated personas may share priors, errors, textual templates and a serving endpoint. Their effective information content is not determined by their nominal count.

B. Graph memory, persona generation and context controls

The intended pipeline builds an event context, generates personas, executes social actions, stores memory and extracts probabilities from a digest. Graph-building and evaluation models were intended to remain fixed across comparisons. That is a design goal rather than a demonstrated property of every original and recovery request. Role names in a configuration are insufficient: the actual endpoint, model identifier, library revision and fallback route should be recorded.

Graph-based retrieval and temporal memory are infrastructure choices. GraphRAG [12] addresses corpus-level question answering through graph indexing and summarization; it does not establish that the present implementation improves forecasting. A functioning Neo4j instance likewise does not prove that agents retrieved relevant memory or that a retrieved fact was historically available at the forecast date. Separating memory construction from the campaign model is useful for experimental control, but introduces another potentially knowledgeable LLM component.

The original context-sanitization protocol removed URLs, rewrote resolution contracts and softened statements about likely outcomes. Those steps may make prompts more consistent; they do not erase model knowledge or establish a historical information barrier. Retrospectively authored dossiers can contain hindsight without explicitly naming the eventual outcome. The term gold-standard context is therefore withdrawn as a validation claim.

C. Why the original inferential arguments are not retained

A sample mean is a valid descriptive statistic even when squared errors are not normally distributed. Distributional assumptions matter for inference about a population, not for reporting the mean of an observed dataset. The earlier assertion that empirical means are statistically invalid is withdrawn.

For paired losses $d_e = L_A(e) - L_B(e)$, the familiar statistic

$$t_d = \frac{\bar{d}}{s_d/\sqrt{E}}$$

uses event pairs, not agents, as observations. It does not address a shared-reference design defect or evaluator leakage. Calling it a Diebold–Mariano statistic without specifying the forecast sequence and long-run variance does not establish time-series validity. In this catalogue, apparently separate questions can share one underlying event, particularly the 2024 US election. An arbitrary event order is not a justified autocorrelation structure.

Observed power computed from the measured effect size largely restates the same test statistic. The earlier combination of incompatible power formulas, an asserted 92-event requirement, and agent-level binomial margins of error is not retained. The illustrative independent-pair normal approximation,

$$E_{\text{approx}} = \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d_z} \right)^2,$$

would give approximately 32 independent pairs for $d_z = 0.5$, two-sided $\alpha = 0.05$ and target power 0.8. This arithmetic is not a sample-size recommendation for ECN-BENCH: it omits event dependence, run variability, selection and uncertainty in the assumed effect. No further runs were selected or scheduled using it.

The old pooled accuracy test has a separate problem. Four campaigns answering the same question do not supply four independent event outcomes, and a tie-aware total such as 65.5 is not an integer binomial count. Rounding it upward does not produce an exact Poisson-binomial test for the reported estimand. Its very small numerical value is withdrawn, not merely re-labelled exploratory.

D. The implemented event-cluster bootstrap

The current computation first averages each event’s contrast across the four fixed campaigns, then resamples events. It uses 100,000 percentile draws with generator seed 7. The US-election sensitivity instead resamples 15 clusters containing 20 events, preserving equal-event weighting when cluster sizes differ. Repeated readings and campaigns belonging to the same sampled event remain together. Resampling agents or evaluator rows independently would manufacture replication.

These intervals quantify variability of the observed event-level contrasts under the chosen resampling scheme. They do not include an unobserved distribution of simulation seeds, future questions, provider configurations or unseen evaluators. Grouping six election questions is a sensitivity analysis, not proof that every remaining event is independent. Neither an interval including zero nor a non-significant comparison proves equivalence.

E. Scoring, ties, sharpness and calibration

A uniform forecast is included in all score summaries. For a binary event that resolves yes, (0.5, 0.5) scores 0.5, whereas (0, 1) scores 2. A reduction in confidence can therefore improve or worsen Brier depending on the outcome and the starting distribution. There is no general rule that a flatter forecast mechanically increases loss relative to another forecast.

Sharpness is a property of the predictive distribution without reference to the realized outcome. The probability assigned to the outcome is not a definition of sharpness. Modal confidence is the maximum forecast probability; directional accuracy asks whether its maximizers include the outcome, with fractional tie credit. The regenerated confidence bins use these two quantities consistently. They do not bin on the known outcome’s probability and then claim to measure top-label calibration.

An empirical reliability diagram with only a few events per bin is descriptive. It cannot establish that a model is calibrated on a future population, and a cell with two successes in two events is not proof of perfect calibration. No calibration transformation was fitted on these same outcomes and then presented as out-of-sample performance.

F. Reference scores, format effects and mathematical coupling

A uniform Brier reference depends on the option count K . Normalizing each event before averaging changes its relative weight compared with an unnormalised mean. The report retains the original BSS definition as a protocol option, but does not call uniform probability a competitive baseline or describe BSS as eliminating all difficulty differences between binary and multicategory questions.

The earlier difficulty–lift plot contains a shared term: its axes are A and $A - B$, where $A = BS_A$ and $B = BS_B$. In general,

$$\text{Corr}(A, A - B) = \frac{\text{Var}(A) - \text{Cov}(A, B)}{\sigma_A \sqrt{\text{Var}(A) + \text{Var}(B) - 2 \text{Cov}(A, B)}}.$$

Even if A and B are independent, this becomes $\sigma_A / \sqrt{\sigma_A^2 + \sigma_B^2}$, which can be large. That coupling does not prove that every observed association is meaningless; it shows why a large correlation alone is not evidence of an informative event-selection rule. A weak correlation between A and B in a small selected sample also does not prove that baseline difficulty has no predictive value.

The historical SBE/SDE labels classify observed score differences after the fact. They are retained in the figure register as names used during investigation, not as independently validated event classes. Selecting events on low BS_A or large $BS_A - BS_B$ cannot then be used as independent evidence that the same score-defined class benefits from simulation.

G. Aggregation budgets and the single-agent estimand

A mean probability vector is a forecast ensemble. Its Brier score differs from the mean score of its members by the mean squared vector dispersion. The report exposes both quantities rather than describing repeated reads as a way to recover a uniquely true swarm forecast. Exact vector agreement is strict, while a within-unit score range grows with the number of available reads; neither supplies a universal measurement threshold.

The existing CoT experiment supplied a dossier and an injection directly, not a newly executed retrieval pipeline. Its three calls use a temperature-zero setting followed by two temperature-0.7 settings. Mean individual-call score estimates the observed mixture of settings, while the three-vector ensemble is a different forecast with a different budget. The two nominally matched model arms are not verified identical-serving comparisons. Without a single-agent C arm, the deliberation-by-input interaction is not identified.

APPENDIX D

EXTENDED SOURCE-CODE AND EXTRACTION AUDIT

A. *The micro-question channel*

The benchmark’s epistemic-mapping proposal asks event-specific micro-questions in addition to a macro forecast. A response contains a dominant tag and a short rationale intended to connect transcript content to a granular event driver. This channel is distinct from both the categorical forecast and the seven rubric dimensions; correctness in one field does not validate the others.

The historical manuscript described widespread missing mappings, often stored as N/A with a missing-response rationale. Such a record can result from a validator filling an omitted field rather than a model explicitly returning that judgment. Missingness should be counted by the actual keys and responses in a specified file. The earlier expression $3 \times 270 \times 3 = 2430$ is not a valid count for three micro-questions across 270 already model-indexed units: it counts the campaigns twice. No current result relies on that aggregate.

The earlier 63-run pilot inventory distinguished provider-style identifiers from local served-model names. Naming is suggestive, not a verified endpoint trace, and those pilot runs change model, prompt, serving and task configuration together. Their reported local/cloud split is a historical troubleshooting observation, not a controlled experiment establishing a deployment-location effect.

B. *Schema bypass and runtime provenance*

A client branch tests model-name substrings including `r1` and `reasoning`. The configured original evaluator name matches that test. The source also contains a local-quantization branch. These are inspectable control paths, not behavioral hypotheses about a model’s intelligence.

The snapshots are not interchangeable. The newer client in `07_pipeline/backend/app/utils/llm_client.py` and the deployment snapshot in `09_audit/source_snapshots/hpc_llm_client.py` document differing constraint and fallback policies. Full request logs and a serving-revision ledger would be needed to assign a precise policy to every historical call. It is therefore incorrect to infer from one current source snapshot that every July request had exactly the same accepted response format.

JSON syntax, nested schema membership and semantic fidelity are separate properties. A schema can require probabilities for all options and still admit a uniform vector. Constraints cannot prohibit hedging unless a task-specific rule explicitly does so, and prohibiting uniform probability would not make the forecast truthful. The original assertion that enforcement forecloses a uniform hedge is withdrawn. The observed J/A difference is a configuration contrast, not a verified intervention on this branch alone.

C. *Retry, validation and silent defaults*

The probability evaluator historically checked some required mappings before accepting a response, but a missing micro-question could be backfilled after that check. A top-level successful call therefore does not imply a complete semantic response. A retry may also alter temperature or feedback, so a final accepted vector need not be a sample from the nominal first-attempt policy.

The older `hpc_evaluator.py` normalization path coerces or drops some invalid values and can return uniform probability when matched mass is non-positive. The newer source rejects that case. The archived snapshot and its SHA-256 make this distinction independently inspectable; neither establishes which individual historical uniform vectors were produced by normalization rather than by the model.

A robust record should retain raw output, the parsed object, option matching, pre-normalization mass, validation exceptions, every retry and the final decision. Final vectors and completion flags alone cannot identify all failure origins. This is why the paper calls near-uniformity an output diagnostic rather than declaring every such unit a failed forecast.

D. Separating predictions from derived metrics

A re-evaluation utility can replace probabilities while leaving stored Brier, ranked probability score, directional accuracy and calibration fields untouched. Such a file can be syntactically valid and scientifically inconsistent. The historical reconciliation step separated evaluator datasets rather than overwriting one pass with another. The current analysis goes further by recomputing the metrics used in the paper directly from each vector.

A scalar must be stored with the hash or version of the prediction from which it was derived. Catalogue identity likewise requires more than an event ID. Two questions may share a yes/no option set and the same recorded outcome, so option/outcome validation cannot catch every topic collision. The surviving Qwen2.5-14B T1 mismatch is a concrete example.

E. Pairing after recovery

The original local evaluator cannot have evaluated newly generated recovery evidence before that evidence existed. Llama’s replaced units and Qwen2.5-14B’s repaired rows are therefore not included as J observations merely because their filename is `event_results.json`. The primary paired audit is restricted to the 132 B/C units whose historical configurations read matching evidence, questions, ordered options and outcomes.

Those checks establish equality of the retained inputs, not equality of complete HTTP requests, micro-question prompts, retry histories or generation settings. The evaluator names are archived execution identifiers; the report does not guarantee current provider routing or weight identity. A repeat call with the same configured seed can still differ under a changed or nondeterministic backend.

F. Existing transfer and truncation controls

The transfer experiment uses one campaign’s evidence and different deployed evaluators under a simplified prompt. Its parsed-response denominators differ. It cannot be treated as a complete-case success rate over all attempted calls, nor as an isolated test of model size. The production-prompt control is useful but does not retroactively match every other configuration.

The truncation grid alters retained content, action selection and speaker coverage together with length. Requested enforcement is not confirmed strict-schema acceptance for every response. The 238 complete off/on pairs are nested within events and lengths, not independent binomial trials. The absence of a clear pooled effect in this proxy does not prove that constraints never matter or identify the original J mechanism. These existing probes are fully reported without adding evaluator calls.

APPENDIX E

RUN INTEGRITY, PERSONA REALIZATION AND EVIDENCE COVERAGE

A. How the evidence categories differ

This appendix restores the detailed operational context behind the short-form findings. It separates four kinds of evidence: frozen vectors reanalysed by executable code; retained source and logs that document a failure path; historical summary scans whose full input collection no longer survives; and mechanistic interpretations that remain untested. Only the first supports the regenerated forecast tables. Historical scan tables are retained as labeled records, not promoted to new matched-campaign results.

B. Persona pools versus configured population

A configured population is an upper bound on the number of distinct persona descriptions, not a measured cognitive sample size. The original pre-repair audit counted persona strings, fallback biographies and literal four-axis labels. Its table is retained below because the failure mode is important: a nominal 300-agent run can be populated largely by repeated templates.

The pre-repair Llama audit describes repeated event-independent archetypes following an endpoint failure. Qwen2.5-7B supplied fewer distinct persona strings than configured agents, while Mistral more often emitted the specified literal axis labels. These are observable formatting and duplication properties. A unique string need not express an independent viewpoint, and a missing literal label need not imply that the corresponding concept is absent. Neither measure supplies a population margin of error.

TABLE XI

HISTORICAL PRE-REPAIR PERSONA-STRING AUDIT OF THE THREE ORIGINAL CAMPAIGNS. LITERAL TEMPLATE COMPLIANCE AND DISTINCT STRINGS DO NOT VALIDATE COGNITIVE DIVERSITY. THE LLAMA ROW INCLUDES SUBSEQUENTLY REPLACED OUTAGE RECORDS; THESE ARE NOT STATISTICS OF ITS RECOVERED CAMPAIGN.

Campaign	Distinct personas / 300 min / median / max	Boilerplate bio	4-axis compliance	Units $\leq$ 20 distinct	
Mistral-7B-Instruct-v0.1	295.6	263/298/300	0.7%	97.7%	0/90
Llama-3.1-8B-AWQ	71.6	11/11/271	79.0%	16.2%	69/90
Qwen2.5-7B-GGUF	38.0	19/40/57	92.0%	1.7%	9/90

C. Outage, partial execution and misleading completion flags

The run-integrity investigation found a temporal break in the original Llama execution sequence: repeated connection failures, a fallback persona library, few organic actions and missing telemetry interviews. The six unaffected events retained in the J/A audit are C3 and S2–S6. Later recovery replaced 24 event records; the historical shorthand of 23 outage events should not be confused with the size of that replacement set, which also includes a transitional/catalogue correction history.

A successful process exit or an orchestrator status of completed does not establish successful agent inference. The affected historical records can carry **error: null** and no evaluator-fallback flag while their simulation logs contain repeated endpoint failures. This does not imply that the evaluator itself failed to make a call: the forecast stage may successfully score an error-filled digest of an unsuccessful simulation.

The distinction matters for units such as the historical Llama C10/B record. The original audit describes an evidence field dominated by connection errors and repeated question/injection lines, yet accompanied by probability and rubric outputs. The scientific defect is that a plausible downstream response masks absent upstream discussion. Rubric scores assigned to such material are not evidence that deliberation occurred.

The old all-campaign action inventory includes parameter sweeps and pre-repair runs. The narrower post-repair scan below is a different collection. Neither can be substituted for the other or used as a controlled model comparison; in particular, the original ranking changed when the execution collection changed.

TABLE XII

REPORTED POST-REPAIR ACTION-COUNT SCAN FOR THE THREE 30-EVENT EXECUTION COLLECTIONS. COUNTS DESCRIBE STORED ACTION TYPES, NOT SEMANTIC QUALITY; THESE HISTORICAL SCAN TOTALS ARE NOT REGENERATED BY THE PROBABILITY REANALYSIS. TELEMETRY INTERVIEWS ARE NOT SUBSTANTIVE SOCIAL POSTS.

Campaign	create_post	create_comment	follow	like_post	like_comment	interview
Llama-3.1-8B-AWQ	9,074	6,454	5,046	4,046	2,044	60,000
Mistral-7B-Instruct-v0.1	3,545	1,648	576	1,618	139	60,000
Qwen2.5-7B-GGUF	281	104	1	14	1	60,000

D. The first recovery merge was also defective

The first attempted recovery rebuilt records from configuration keys that did not contain the expected question/evidence fields. The recorded problem was not a small numerical discrepancy: evidence could be empty, a condition-insensitive merge could overwrite the shared A reference, and one source could populate nominally distinct evaluator files. Those mistakes created apparent differences between conditions and apparent agreement between evaluators for the wrong reasons.

The corrected recovery rebuilt evidence from available logs/actions, restored the shared reference instead of re-eliciting it, and separated evaluator outputs. It also used a hosted Llama endpoint rather than the original local AWQ deployment. The repair is therefore a replacement execution under a changed serving configuration, not proof that the original local run can be replayed exactly.

A generation budget that succeeds on empty evidence may fail on a full transcript. The historical recovery record reports null completions and increased completion budgets during rescoring. That change is part of the extraction configuration; it must not be hidden behind a single evaluator-model label. The revision adds no recovery execution of its own.

E. Catalogue collisions and the remaining T1 mismatch

The stale catalogue reused short event IDs for different questions. A downstream micro-question collision and an upstream wrong-topic simulation are different defects. Correcting a micro-question label does not repair a discussion generated for the wrong event, and repairing a discussion does not automatically repair the separate injection bank.

Qwen2.5-14B previously had six wrong-topic events replaced. During restoration of the full per-event tables, a stricter cross-campaign question check identifies a remaining T1 mismatch in all three primary conditions: the other campaigns ask whether the UAW reaches an agreement with the Big Three automakers before November 1, 2023, while Qwen2.5-14B asks about a Fed rate cut in November 2024. The yes/no options and recorded outcome coincide. This is precisely the kind of collision that a probability-simplex or outcome-key check cannot detect.

The full numerical appendix retains the stored T1 rows so the defect is visible, but excludes T1 across all four campaigns from its 29-event aligned summaries and confidence bins. The main 20-event subset already excludes T1, so its estimates are unchanged. No question, vector or simulated evidence was edited to conceal the mismatch. Recovering T1 with another run is not necessary to state the present audit results and has not been attempted.

Qwen2.5-14B C9/C separately retains an outage-like integrity concern. The main paper reports an input/gap sensitivity excluding C9 along with C6, T6 and T8. Alignment of the remaining 29 questions is not certification of simulation health.

F. What survives when raw traces are missing

Qwen2.5-14B’s six recovered events and an original C15 fragment retain raw artifacts. Original raw traces for 21 other events are unavailable following scratch-storage deletion. Scored evidence fields survive, so probability reanalysis and digest inspection remain possible. Persona and database scans requiring the lost originals cannot be reproduced across that entire campaign.

The original Qwen2.5-14B all-campaign SQL scan predates the six question repairs. Its counts include wrong-topic executions; the affected count is six events, not six database files. Event, condition, platform database, request and evaluator-read counts must not be interchanged. The original 180-database inventory is retained as a historical inventory, not reconstructed as if it were the corrected 120 B/C platform databases.

G. Sampled evidence is not the complete conversation

The scored evidence is a digest: a log tail and a representative-action block. A historical sampler ranks actions by a fixed engagement heuristic, selects up to 60 and orders those selections by round. The cap limits what the evaluator sees; it does not make all evidence equally informative or guarantee that every unit reaches the cap. Sparse runs can include telemetry scaffolding instead of substantive utterances.

Probe lines such as a belief check with a missing value are instrumentation artifacts, not independent arguments from a person. Counting distinct speaker labels without examining message content can therefore overstate conversational diversity. Conversely, a small text sample does not establish that the entire underlying simulation was inactive.

TABLE XIII

HISTORICAL DIGEST AUDIT OF THE THEN-CURRENT SCORED EVIDENCE: PROPORTION OF SAMPLED LINES IDENTIFIED AS TELEMETRY PROBES. THESE FIGURES ARE RETAINED FROM THE ORIGINAL AUDIT RECORD, NOT RECOMPUTED AS PART OF THE PRESENT PROBABILITY ANALYSIS. INSTRUMENTATION CONTENT IS NOT AUTOMATICALLY OUTCOME LEAKAGE.

Campaign	Probe share (mean)	Probe share (median)	Distinct utterances
Llama-3.1-8B	0.0%	0.0%	56
Mistral-7B-Instruct-v0.1	0.0%	0.0%	41.5
Qwen2.5-7B-GGUF	86.0%	86.7%	8
Qwen2.5-14B-AWQ	72.0%	90.0%	6

Telemetry in the digest is not automatically ground-truth leakage. A valid agent probability probe may be a legitimate part of the input, whereas an invalid placeholder is missing measurement. The methodological problem is that the pipeline did not isolate these channels or consistently distinguish them in its final record. Campaign-level differences in probe share are confounded with model and serving identity, so a pooled correlation cannot identify their effect on accuracy.

Option names can also occur as persona display names. Seeing a candidate’s name in a political discussion is not by itself proof that the outcome label leaked: an incorrect option can appear the same way, and all options may be known before resolution. An outcome-aware selection of names would be more serious, but that provenance is not established by an exact string match. The corrected interpretation is a possible cue channel requiring an outcome-blind check, not a demonstrated source of the forecast’s accuracy.

H. Two qualitative event readings

The original close reading contrasted Llama C11, the Nobel Peace Prize question, with S6, the Austrian election question. Those discussions are retained as qualitative diagnostic examples, not a matched causal experiment: S6 is among the unchanged local units and C11 belongs to the recovered hosted execution, so they do not share one serving history.

For C11, the audit described generic social advocacy plus an injection favoring atomic-bomb-survivor organizations. A resulting forecast can react to that cue without demonstrating that agent-to-agent deliberation discovered the outcome. A persona name matching an option is a possible additional prompt cue, not conclusive proof of outcome contamination.

For S6, the sampled discussion was largely about sustainability and other off-topic themes rather than the Austrian election. Fluent agreement in such a sample is not evidence of task-relevant consensus. An extractor assigning mass to a residual option can reflect its interpretation of that sample, its own prior or both.

These examples motivate a distinction between generating conversation and delivering relevant evidence to the extractor. They do not estimate the prevalence of either pattern or locate a general mechanism. Small score differences between a CoT call and a swarm on one chosen event are not equivalence tests. The full per-event numerical tables make the current vector-derived values available without presenting the selected anecdotes as confirmatory results.

I. Rubric repeatability and construct validity

The historical rubric compares repeated calls of a deployed evaluator, not independent human raters. Cohen’s kappa describes label agreement after its marginal adjustment and depends on label prevalence. A low value is a repeatability warning; a high value would still not demonstrate that the labels measure polarization, herding or deliberation quality.

TABLE XIV

HISTORICAL REPEATED-CALL RUBRIC AGREEMENT VALUES. THE OPERATIONAL 0.8 CUTOFF WAS A PROJECT RULE, NOT A UNIVERSAL VALIDITY THRESHOLD. CAMPAIGN AND EVALUATOR REVISIONS ARE NOT MATCHED; THESE FIGURES DO NOT RANK MODEL CAPACITY.

Dimension	Llama-3.1-8B	Mistral-7B	Qwen2.5-7B
prediction_accuracy	0.321	0.182	0.088
polarization	0.241	0.167	0.122
deliberation_quality	0.209	0.228	0.062
herd_effect	0.206	0.113	0.053
convergence	0.245	0.139	0.212
susceptibility	0.322	0.245	0.033
information_diversity	0.228	0.130	0.103
Max κ	0.322	0.245	0.212
Dropped dims	7	7	7

The project applied a 0.8 operational threshold and dropped unstable dimensions. That threshold is not a universal scientific boundary, and moving to a finer scale does not guarantee greater validity. No EPI composite is restored as an empirical model ranking. The proposed dimension definitions remain available in the technical specification so that future validation can examine them rather than assuming them correct.

J. Telemetry placeholders and the meaning of convergence

The historical JSD audit found repeated initialization-like values and constant series. A non-increasing series passes a non-strict monotonicity test even when it never changes. It follows that a count of monotonic runs cannot establish convergence without verifying successful probes and understanding what distributions were compared.

TABLE XV

HISTORICAL TELEMETRY INTEGRITY SUMMARY, BEFORE RECOVERY: 60 B/C UNITS AND 600 CHECKPOINTS PER CAMPAIGN. THE LAST COLUMN COUNTS DECREASING TRANSITIONS OUT OF 540, NOT CONVERGING RUNS. CONSTANT INITIALIZATION VALUES CANNOT ESTABLISH CONVERGENCE.

Model	Mean JSD	Pipeline “monotonic”	Placeholder values	Constant series	Strictly decreasing
Llama-3.1-8B-AWQ	0.734	48	526/600 (87.7%)	48/60	29
Mistral-7B-Instruct-v0.1	0.637	0	244/600 (40.7%)	0/60	133
Qwen2.5-7B-GGUF	0.628	0	166/600 (27.7%)	0/60	113

“Placeholder values” counts checkpoint readings exactly equal to 0.758277, the constant the telemetry emits when no probed distribution is available. “Constant series” counts units whose ten checkpoint values are all identical. “Strictly decreasing” counts round-to-round transitions with a JSD decrease exceeding the pipeline tolerance $\epsilon = 0.002$, out of 540 transitions per model.

The pipeline’s JSD construction also needs to be distinguished from inter-agent disagreement. A distance between an aggregated histogram and a fixed uniform reference is not the same as pairwise belief dispersion, and neither is automatically a distance to a correct consensus. A decline can reflect a changing reference relationship, aggregation or placeholder behavior rather than increasing agreement among agents. A valid DeGroot interpretation requires additional structure not measured here.

K. Topology, behavior and limits of interpretation

The historical run manifest records an unspecified default degree distribution rather than a measured power law. The previously asserted exponent is withdrawn. A directed complete graph has at most $N(N - 1)$ non-self edges; that bound is not the realized edge count, message volume or memory load in this sparse execution.

Follows, replies, likes and dislikes describe selected tool actions. They do not identify antagonism, reflective cognition or a causal epistemic mechanism without content validation and controlled comparison. A low follow count does not make topology irrelevant, because an initial graph or recommendation mechanism may still expose posts. Nor does agreement across campaigns prove shared memorization or social amplification.

Interventions such as forced posting, reply requirements or diversity penalties would change the simulated process. They may be useful future experiments but are not repairs that can be assumed to preserve the original estimand. No such intervention was run in this revision.

L. Temporal exposure and the retained event subsets

A low baseline Brier score can result from an easy dossier, base rates, fortunate confidence, genuine inference or prior knowledge. Thresholding that score does not identify training contamination. It also selects on a component of the lift statistic, so a lift reversal after removing low-baseline-score events is partly a consequence of the selection rule.

Release dates are a more externally defined descriptive stratifier, but the forecast is produced by the whole pipeline. Later extractors may know outcomes, and retrospectively authored inputs can expose them. A post-release label for a campaign checkpoint is therefore not a certificate that the observed pipeline forecast was made under genuine future uncertainty.

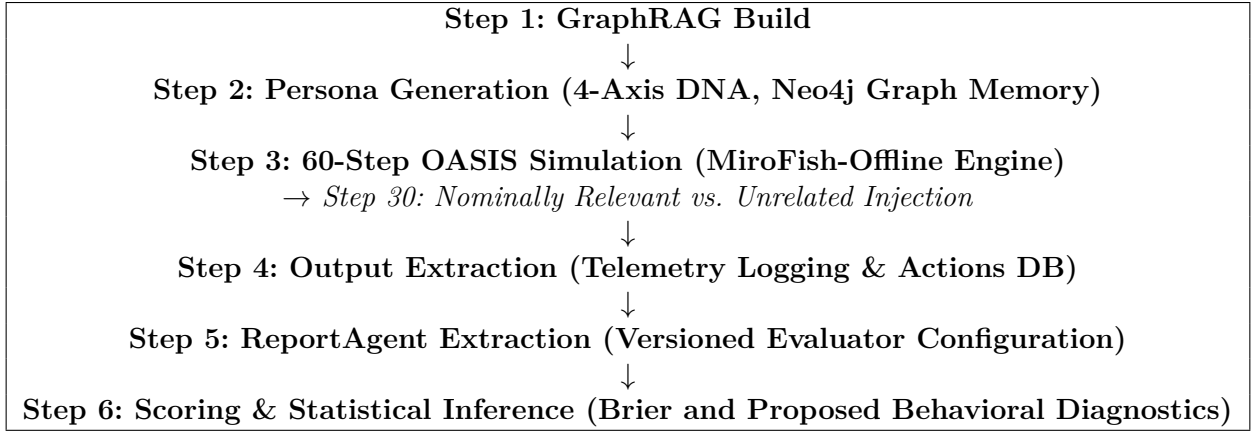


Fig. 2. The ECN-BENCH Multi-Agent Social Simulation Pipeline.

The retained 20-event set is listed in the main analysis. Six entries share the US election and are grouped or omitted in sensitivity checks. Three contain wrong-topic injections and C9 carries a further integrity concern. The unchanged primary comparison should be read with all these qualifications, rather than selecting whichever evaluator or subset gives the strongest interval.

APPENDIX F

SYSTEM ARCHITECTURE AND ENGINEERING (MIROFISH-OFFLINE)

The historical execution stack was designed for restricted-network HPC environments. The architecture below describes the intended pipeline and engineering revisions, not a claim that every campaign completed all steps successfully. Original and recovered runs do not share one immutable stack.

A. The Local Serving Stack

The project’s original compute jobs used restricted-network nodes, motivating local services. This is a constraint of the recorded environment, not a claim that every university cluster prohibits outbound access. Hosted recovery and post-campaign evaluator calls are separately identified. The intended local components were:

- **Primary Inference Server:** vLLM [13] is utilized to host open-weight models (e.g., Qwen2.5-72B-Instruct, Llama-3-70B-Instruct). vLLM’s continuous batching and PagedAttention [13] support batched serving.
- **Embedding Server:** An Infinity or Ollama background server hosting `nomic-embed-text` handles embedding queries for vector similarity.
- **Graph Engine:** A self-hosted Neo4j Community Edition instance coupled with the Graphiti context library manages the relational and temporal state of the swarm memory.

B. HPC Scalability Migration (Architecture v5.0 - v5.10)

Scaling and concurrent telemetry exposed routing, container and storage bottlenecks. The following changes document the engineering response. They reduced specific failure risks; the later outage shows that they did not establish absolute stability.

1) *Hybrid LLM Routing (v5.0):* The original routing experiment separated chat generation from embedding service load. Primary interaction could use vLLM on port 8000 while embedding requests used a separate service on port 11434. This is a workload-isolation choice, not a general claim that Ollama lacks batching or cannot serve concurrent requests.

2) *Multi-GPU Server Isolation (v5.5)*: To support 100-agent simultaneous telemetry probes without stalling the social simulation, a 4-GPU isolation protocol was established:

- **Benchmark Server (GPUs 0,1)**: Handles active agent interactions (Post/Reply/Like) on Port 8000.
- **Evaluator/Telemetry Server (GPUs 2,3)**: Dedicated to probability probing and final scoring on Port 8001.

This separation reduces direct resource contention; it does not guarantee immunity to timeouts, queue saturation or shared-node failures. Actual GPU assignments must be read from each job rather than inferred from this reference layout.

3) *Apptainer Containerization (v5.6 - v5.7)*: The project used Apptainer/Singularity containers, including a recorded vLLM 0.6.3-based image, to reduce host-environment variation. Experimental bind mounts replaced selected Python packages when the image lacked support for a checkpoint. The illustrative mount below is an engineering record, not a guaranteed reproducible environment: a changed library tree alters the software revision even when the container image name is unchanged.

$$\begin{aligned} \text{Bind Mount Target: } & - B \text{ /scratch/waves/transformers} \\ & : /usr/local/lib/python3.12/ \\ & \text{dist-packages/transformers} \end{aligned} \tag{20}$$

This replaces the immutable container libraries at runtime with development versions, adjusting for Python 3.12 container environments while the host runs Python 3.11.

4) *NFS Database Concurrency and Waves NVMe Migration*: The project encountered SQLite lock errors on network-backed storage during concurrent activity. Two recorded mitigations were a longer connection timeout and relocation of active data to node-local scratch. These can reduce contention but do not eliminate all locking errors or guarantee transaction safety:

- 1) **The Connection Lock Protocol**: All SQLite connectors enforce a strict 60-second queue timeout:

```
sqlite3.connect(database_path, timeout=60)
```

- 2) **Waves NVMe Relocation**: Neo4j graph data and active .db files are symlinked directly onto local high-I/O scratch spaces (/scratch/waves/ NVMe drives), reducing network-storage latency without guaranteeing absence of lock contention.

C. Reproducibility costs of mutable environments

A container digest, bind-mounted package tree, model weights, tokenizer and serving flags jointly define the execution environment. A container tag alone is insufficient. Recovery via a hosted endpoint can restore usable evidence without restoring the original hardware, numerical precision or routing behavior. The release therefore separates analysis reproducibility from replay of the complete historical system.

Scratch relocation also creates a retention risk. The missing Qwen2.5-14B originals show that successful execution is not a preservation strategy. Source data should be exported before ephemeral storage expires; manifests and checksums should describe what was actually exported, including partial coverage. This report does not claim that missing raw artifacts have been reconstructed.

APPENDIX G

COMPLETE SCALING AND COMPUTATIONAL-COST RECORD

The original scaling campaign is retained as an engineering exploration rather than removed from the report. It examined two anchor questions, C1 and C9, using one run per configuration/condition. No new sweep was run for this revision. The original numerical matrices below are retained as historical execution records; the current probability-analysis scripts do not claim to regenerate their full execution history.

TABLE XVI
HISTORICAL SINGLE-RUN POPULATION SWEEP AT 60 ROUNDS. EACH CELL IS AN OBSERVED EXECUTION RECORD, NOT A POPULATION ESTIMATE; NO REPLICATED UNCERTAINTY IS AVAILABLE.

N (agents)	Event C1: US Election			Event C9: UK Trade		
	Cond A	Cond B	Cond C	Cond A	Cond B	Cond C
30	0.3368	0.5300	0.4200	0.2150	0.5952	0.6069
100	0.3368	0.3958	0.5958	0.2150	0.6971	0.6329
300	0.3368	0.5350	0.6800	0.2150	0.7630	0.8197
1000	0.3368	0.9533	0.9000	0.2150	0.7264	0.6417
3000	0.3368	0.6133	0.7911	0.2150	0.7885	0.6095

A. Population sweep

At 60 rounds, the population settings were 30, 100, 300, 1,000 and 3,000 agents. The table supports a non-monotone description, not a fitted scaling law. On C1 the lowest observed B score is at 100 agents; on C9 it is at 30, not in the previously claimed 100–300 optimal zone. At 1,000 agents, C9 scores below its 300-agent value, while C1 scores above it. Those facts contradict a universal monotonic degradation story. The theoretical upper bound $N(N - 1)$ on directed non-self links is not the realized number of graph edges or messages. A complete graph at 3,000 agents would have many potential links, but the deployed interaction policy is not a complete communication network. Neither memory saturation nor collapsing signal-to-noise was directly measured by these Brier cells. A uniform distribution also maximizes, rather than minimizes, Shannon entropy, so an earlier low-entropy uniform-weight argument is mathematically incorrect.

B. Round sweep

At 300 agents, the recorded round settings were 30, 60 and 90. C1 Brier falls from 0.7200 to 0.5350 and then rises to 0.9983. C9 changes from 0.7721 to 0.7630 to 0.7850. The much smaller C9 differences cannot be treated as independent confirmation of a sharp collapse boundary. With one realization per cell, between-run variability is unknown. Sixty rounds yield a lower observed Brier than the other two round settings on these

TABLE XVII
HISTORICAL SINGLE-RUN ROUND SWEEP AT 300 AGENTS. A CHANGE IN BRIER DOES NOT ESTABLISH CONVERGENCE OR LOCATE AN ECHO-CHAMBER THRESHOLD.

R (rounds)	Event C1: US Election			Event C9: UK Trade		
	Cond A	Cond B	Cond C	Cond A	Cond B	Cond C
30	0.3368	0.7200	0.6283	0.2150	0.7721	0.7386
60	0.3368	0.5350	0.6800	0.2150	0.7630	0.8197
90	0.3368	0.9983	0.6958	0.2150	0.7850	0.8401

two anchors. That is not a proof that agent beliefs converge by rounds 40–50 or that an echo chamber starts at rounds 70–80. The telemetry integrity issues independently prevent that interpretation. The negative A-to-B contrast on C1 is not a positive lift merely because B improves between round settings.

C. Why 300 agents and 60 rounds were selected

The production configuration is a pragmatic historical choice balancing execution cost, space for persona diversity and the observed anchor-event scores. It is not an empirically established optimum. The claim that 300 generated agents provide a binomial 95% margin of error of 0.056 assumes independent observations of a common population quantity that these correlated language-model personas do not supply. That agent count also has no direct role in the event-level bootstrap denominator.

Distinct persona strings must be measured, and four conceptual axes do not partition 300 agents into four independent samples of 75. The persona audit shows why nominal capacity and realized variety differ. A larger configured population can increase resource demand without increasing independent task-relevant information.

The configuration should therefore be interpreted as an execution parameter held nominally fixed in the main campaigns. A new choice would require a prespecified utility/cost objective and replicated measurements under the intended workload. This report neither recommends another production setting nor launches the simulations needed to evaluate one.

D. Cross-model anchor checks

The anchor comparisons retain model-specific reference values and unbalanced coverage. Qwen2.5-14B has only C1 in this particular matrix; that is not its coverage in the separate recovered 30-ID campaign. A missing C9 cell is not zero and should not be filled from a different campaign. The positive Qwen2.5-7B A-to-B contrast

TABLE XVIII

HISTORICAL ANCHOR-EVENT CAMPAIGN COMPARISON AT THE SELECTED CONFIGURATION. MODEL, SERVING, PRECISION AND REFERENCE CONSTRUCTION ARE NOT MATCHED. THESE ARE NOT A VALIDATED RANKING OR A UNIVERSAL OPTIMUM.

Model	Role	Event C1: US Election			Event C9: UK Trade		
		Cond A	Cond B	Cond C	Cond A	Cond B	Cond C
Llama-3.1-8B-AWQ	Reference	0.3368	0.5350	0.6800	0.2150	0.7630	0.8197
Qwen2.5-7B	Validation	0.5600	0.4450	0.6217	0.2150	0.6080	0.7209
Qwen2.5-14B	Validation	0.5600	0.5978	0.5958	—	—	—
Mistral-7B-Instruct-v0.1	Validation	0.5600	0.6600	0.6333	0.2150	0.8286	1.1037

on C1 and negative contrasts in other cells are descriptive. They do not demonstrate that one architecture benefits from deliberation and another saturates before 60 rounds. Model generation, serving and precision differ, and no replicated factorial experiment separates those factors.

E. Runtime and resource accounting

Each recorded event job contains setup, reference/persona work, simulation B, simulation C and evaluator-related work. The original table separates phase durations and total scheduler wall-clock time. A wall-clock

TABLE XIX

HISTORICAL SINGLE-EVENT C1 WALL-CLOCK RECORD ON THE RECORDED A100 DEPLOYMENT. JOB DURATION INCLUDES SETUP AND TEARDOWN; IT IS NOT GPU-HOURS, TOKEN COST OR A THROUGHPUT GUARANTEE.

Model	Phase A (Persona Gen+Eval)	Sim B (With-Sim)	Sim C (Null)	Wall-Clock Total
Llama-3.1-8B-AWQ	3.8 min (229s)	35.1 min (2104s)	31.4 min (1883s)	77.6 min (4655s)
Qwen2.5-7B	2.7 min (161s)	47.9 min (2874s)	18.4 min (1103s)	77.1 min (4624s)
Qwen2.5-14B	2.7 min (160s)	69.5 min (4168s)	31.5 min (1890s)	114.0 min (6842s)
Mistral-7B-Instruct-v0.1	2.7 min (160s)	33.7 min (2024s)	30.1 min (1806s)	75.4 min (4526s)

Each row reflects a single-event (C1) three-condition run from the July 2026 v523 optimization sweep at $N = 300$, $R = 60$ on

A100. “Wall-Clock Total” is the Slurm job wall time and includes setup/teardown overhead, so phase times need not sum exactly to it. LLM persona generation alone is approximately 49 s (0.8 min); the remainder of Phase A (~ 110 – 180 s) is persona evaluation and scoring.

minute is not a GPU-minute unless the allocation is one GPU for that entire interval. Idle reservation, parallel services, retries and differently sized allocations matter. The phase durations need not sum to the scheduler total, and the table does not establish a general linear cost law or a monetary price. Earlier conversions from these wall-clock values into campaign-wide GPU-hour requirements are not retained.

A configuration with lower observed loss and longer runtime can be preferable under one utility function and worse under another. Even on C1, 60 rounds cannot be called Pareto-dominant over 30 merely because it has lower Brier: the shorter configuration may cost less. No population cost-performance frontier is established by two selected questions with no repeated runs.

F. Hardware routing and deployment failures

The historical log discusses initialization failures on older GPU nodes, context-window overflow, KV-cache allocation errors and persona JSON parsing failures. These are distinct failure classes. A configured maximum context of 4,096 versus a prompt needing more space is an interface-budget problem; increasing it can increase memory demand. A memory-utilization flag is a fraction of a deployment’s available resources, not a portable capacity threshold.

The original troubleshooting settings of 8,192 maximum tokens and memory fractions 0.40 or 0.45 document particular attempts, not guaranteed minimums for other hardware, batch sizes, tokenizers or serving revisions. The recorded A100 scheduling constraint addressed the local deployment; it is not a claim that every older GPU is incapable of every operation required by the pipeline.

Guided-decoding compilation, schema mismatch and invalid unconstrained content also require separate diagnosis. The fact that two small Gemma-family deployments failed does not identify a 15B parameter boundary, an architectural impossibility, or a general incompatibility of all AWQ checkpoints. Successful smaller deployments elsewhere in the same project already make such a universal claim untenable.

G. Revised configuration decision record

TABLE XX

WHAT EACH CONFIGURATION CONSIDERATION CAN AND CANNOT ESTABLISH. THIS REPLACES THE ORIGINAL DECISION MATRIX’S UNVALIDATED PRECISION AND ECHO-CHAMBER LABELS.

Consideration	Observable in this record	Not established
Agent count	Configured slots; recorded persona strings	Independent sample size or cognitive coverage
Round count	Executed horizon and single-run score	Consensus time or collapse threshold
Brier	Loss of the extracted probability vector	Truthfulness of its interpretation of the swarm
Runtime	Deployment-specific elapsed duration	Portable GPU-hours, price or general scaling law
Schema handling	Source branches and some response records	Semantic correctness or universal prevention of uniformity
Model size	Identity of attempted deployments	A sub-15B exclusion rule

APPENDIX H

EXTENDED FOLLOW-UP PROGRAMME

These items restore the original future-work scope while separating proposals from completed experiments. None was executed or scheduled in this revision.

- 1) **Extraction fidelity:** construct outcome-blind reference transcripts or obtain blinded human judgments of what the discussion supports. This would distinguish a different extractor from a more faithful one.
- 2) **Evaluator prior knowledge:** compare question-only, dossier-only and transcript-conditioned extraction with versioned prompts. This requires additional calls, not necessarily new simulations, and would still not by itself certify freedom from leakage.
- 3) **Deliberation contribution:** cross single-agent versus simulation with matched B/C inputs and model/serving conditions. Repeated simulation seeds would be needed to study between-run stability.
- 4) **Input integrity:** version/hash catalogue questions, ordered options, dossiers and injection bodies. Treat topic mismatches as invalid pairings, not as values to repair silently after analysis.
- 5) **Telemetry integrity:** retain successful and failed probe counts, missingness and raw probability vectors. Exclude invalid checkpoints from convergence measurements rather than assigning a numeric default.
- 6) **Persona and graph realization:** measure distinct descriptions, realized topology and task-relevant information flow. Do not equate string diversity with cognitive independence.

- 7) **Scaling:** define a cost/loss objective before comparing replicated settings on new questions. The two existing anchors are exploratory design inputs, not a validation set.
- 8) **Artifact preservation:** export ephemeral storage, preserve raw completions and record accepted schema status, retry history and model/serving revisions.

The first two items are particularly relevant to extending the extraction claim; the third is needed for a stronger claim about deliberation itself. None is required to report the current fixed-evidence configuration observation with its stated limitations. Increasing the number of runs simply to obtain a smaller p -value is not a justified follow-up strategy.

APPENDIX I

COMPLETE EVENT REGISTER AND REGENERATED NUMERICAL TABLES

A. Complete question and outcome register

Recorded outcomes are reproduced for audit, not independently adjudicated. Post-release denotes the retained date-based subset, not freedom from evaluator knowledge. Wrong injection identifies the known stale-bank topic mismatch.

C1: Which party will win the 2024 US Presidential Election?

- Recorded outcome: Republican.
- Ordered options: Republican; Democrat; Other.
- Post-release subset: yes; wrong-topic injection: no.

C2: Who will be the 2024 Democratic presidential nominee?

- Recorded outcome: Harris.
- Ordered options: Biden; Harris; Newsom; Michelle Obama; Other.
- Post-release subset: no; wrong-topic injection: no.

C3: Which party wins the most seats in the 2024 French snap legislative election?

- Recorded outcome: Nouveau Front Populaire.
- Ordered options: Rassemblement National; Nouveau Front Populaire; Ensemble; Other.
- Post-release subset: no; wrong-topic injection: no.

C4: Which party will win the most seats in the 2024 UK general election?

- Recorded outcome: Labour.
- Ordered options: Labour; Conservative; Reform UK; Other.
- Post-release subset: no; wrong-topic injection: no.

C5: What will be the balance of power after the 2024 US election?

- Recorded outcome: Republican President + Republican Congress.
- Ordered options: Republican President + Republican Congress; Republican President + Split Congress; Democratic President + Democratic Congress; Democratic President + Split Congress; Other.
- Post-release subset: yes; wrong-topic injection: no.

C6: Which geographic origin will the next Pope elected in the 2025 Papal Conclave come from?

- Recorded outcome: North American Cardinal.
- Ordered options: Italian / European Cardinal; Latin American Cardinal; African Cardinal; North American Cardinal; Asian or Other Cardinal.
- Post-release subset: yes; wrong-topic injection: yes.

C7: Which country will win the 2024 UEFA European Championship?

- Recorded outcome: Spain.
- Ordered options: Spain; England; France; Germany; Portugal; Other.
- Post-release subset: no; wrong-topic injection: no.

C8: Which AI company will first release a model scoring >90% on MMLU?

- Recorded outcome: OpenAI.
- Ordered options: OpenAI; Google DeepMind; Anthropic; Meta; Other.
- Post-release subset: yes; wrong-topic injection: no.

C9: Which countries will the US agree trade deals with before July 2025?

- Recorded outcome: UK.
- Ordered options: UK; India; Japan; Vietnam; South Korea; EU; Canada; Mexico; Other / None.
- Post-release subset: yes; wrong-topic injection: no.

C10: Who will be the next UK Prime Minister after the 2024 election?

- Recorded outcome: Keir Starmer.
- Ordered options: Keir Starmer; Rishi Sunak; Nigel Farage; Other Conservative; Other.
- Post-release subset: no; wrong-topic injection: no.

C11: Who will win the 2024 Nobel Peace Prize?

- Recorded outcome: Nihon Hidankyo.
- Ordered options: Nihon Hidankyo; Pope Francis; UN Agency; Individual Activist; Other.
- Post-release subset: yes; wrong-topic injection: no.

C12: Which team will win the 2024 NBA Championship?

- Recorded outcome: Boston Celtics.
- Ordered options: Boston Celtics; Oklahoma City Thunder; Denver Nuggets; Minnesota Timberwolves; Other.
- Post-release subset: no; wrong-topic injection: no.

C13: Who will win the 2025 Oscar for Best Picture?

- Recorded outcome: Anora.
- Ordered options: Anora; The Brutalist; Conclave; Emilia Pérez; Wicked; Other.
- Post-release subset: yes; wrong-topic injection: no.

C14: Which party will form the next government after the 2025 German federal election?

- Recorded outcome: CDU/CSU-led coalition.
- Ordered options: CDU/CSU-led coalition; SPD-led coalition; AfD-led coalition; Other / Hung parliament.
- Post-release subset: yes; wrong-topic injection: no.

C15: How many total rate cuts will the Fed deliver in 2024?

- Recorded outcome: 3.
- Ordered options: 0; 1; 2; 3; 4; 5+.
- Post-release subset: yes; wrong-topic injection: no.

S2: Which party will control the US Senate after the 2024 election?

- Recorded outcome: Republican.
- Ordered options: Republican; Democrat.
- Post-release subset: yes; wrong-topic injection: no.

S3: Will Republicans achieve a red sweep in 2024 (Presidency + House + Senate)?

- Recorded outcome: YES.
- Ordered options: YES; NO.
- Post-release subset: yes; wrong-topic injection: no.

S4: Will total 2024 presidential election turnout fall in the 155–160 million bracket?

- Recorded outcome: YES.
- Ordered options: YES; NO.
- Post-release subset: yes; wrong-topic injection: no.

S5: Will the GOP popular vote margin of victory be less than 1.5%?

- Recorded outcome: YES.
- Ordered options: YES; NO.
- Post-release subset: yes; wrong-topic injection: no.

S6: Which party wins the most seats in the 2024 Austrian legislative election?

- Recorded outcome: FPÖ.
- Ordered options: FPÖ; ÖVP; SPÖ; Other.
- Post-release subset: yes; wrong-topic injection: no.

S7: Will OpenAI become a for-profit company in 2025?

- Recorded outcome: YES.
- Ordered options: YES; NO.
- Post-release subset: yes; wrong-topic injection: no.

S9: Will there be a US government shutdown in early 2025?

- Recorded outcome: YES.
- Ordered options: YES; NO.
- Post-release subset: yes; wrong-topic injection: no.

T1: Will the UAW reach a contract agreement with all three Big Three automakers (GM, Ford, and Stellantis) before November 1, 2023?

- Recorded outcome: YES.
- Ordered options: YES; NO.
- Post-release subset: no; wrong-topic injection: yes.

T2: Will there be a US government shutdown before January 1 2025?

- Recorded outcome: NO.
- Ordered options: YES; NO.
- Post-release subset: yes; wrong-topic injection: no.

T3: Which party will win the most votes in the 2024 Taiwanese presidential election?

- Recorded outcome: DPP.
- Ordered options: DPP; KMT; TPP.
- Post-release subset: no; wrong-topic injection: yes.

T4: Who will be elected US House Speaker following Kevin McCarthy's removal in October 2023?

- Recorded outcome: Mike Johnson.
- Ordered options: Mike Johnson; Jim Jordan; Tom Emmer; Steve Scalise; Other.
- Post-release subset: no; wrong-topic injection: yes.

T5: Will AB InBev report a year-over-year decline in US revenue in their Q2 2023 earnings due to the Bud Light boycott?

- Recorded outcome: YES.
- Ordered options: YES; NO.
- Post-release subset: no; wrong-topic injection: yes.

T6: Which party will win the most seats in the 2024 Irish general election?

- Recorded outcome: Fianna Fáil.
- Ordered options: Fianna Fáil; Sinn Féin; Fine Gael; Other.
- Post-release subset: yes; wrong-topic injection: yes.

T8: Which party will win the most seats in the 2025 Canadian federal election?

- Recorded outcome: Liberal.
- Ordered options: Liberal; Conservative; NDP; Bloc Québécois; Other.
- Post-release subset: yes; wrong-topic injection: yes.

T9: Will OpenAI announce it has achieved AGI in 2025?

- Recorded outcome: NO.
- Ordered options: YES; NO.
- Post-release subset: yes; wrong-topic injection: no.

B. Question-aligned summaries and prefix diagnostics

T1 is excluded from every campaign in aggregate comparisons: Qwen2.5-14B T1 asks about a Fed decision rather than the UAW contract question. The options and recorded outcome coincide, so vector validation alone misses it. Per-event tables retain all 30 stored IDs, explicitly including this invalid pairing. The primary 20-event post-release subset never includes T1.

TABLE XXI

THE 29 QUESTION-ALIGNED EVENTS UNDER EVALUATOR A’S AVERAGED VECTOR. T1 IS EXCLUDED ACROSS CAMPAIGNS. THE A-CONDITION COLUMN IS A REREADING OF SHARED REFERENCE EVIDENCE, NOT AN INDEPENDENTLY EXECUTED BASELINE FOR EACH MODEL.

Campaign	BS A	BS B	BS C	Acc. B	Uniform B
Llama-3.1-8B	0.376	0.445	0.536	0.776	0
Mistral-7B	0.293	0.469	0.506	0.724	0
Qwen2.5-7B	0.307	0.299	0.451	0.862	0
Qwen2.5-14B	0.314	0.262	0.369	0.931	0

TABLE XXII

DESCRIPTIVE PREFIX GROUPS UNDER A’S AVERAGED VECTOR, CONDITION B. C/S/T ARE RECORD IDENTIFIERS, NOT A VALIDATED DOMAIN TAXONOMY. CAMPAIGN IDENTITIES AND SERVING REMAIN CONFOUNDED.

Campaign	Prefix	Events	BS B	Acc. B
Llama-3.1-8B	C	15	0.461	0.767
Llama-3.1-8B	S	7	0.591	0.571
Llama-3.1-8B	T	7	0.265	1.000
Mistral-7B	C	15	0.489	0.733
Mistral-7B	S	7	0.447	0.714
Mistral-7B	T	7	0.450	0.714
Qwen2.5-7B	C	15	0.244	0.933
Qwen2.5-7B	S	7	0.474	0.714
Qwen2.5-7B	T	7	0.243	0.857
Qwen2.5-14B	C	15	0.218	0.933
Qwen2.5-14B	S	7	0.305	0.857
Qwen2.5-14B	T	7	0.313	1.000

C. Modal confidence and empirical accuracy

These are top-label confidence bins: confidence is the maximum forecast probability, and accuracy uses the same tie credit as the main analysis. Confidence is not the probability assigned to the known ground-truth option. Small bin counts, reuse of events and retrospective evaluator exposure preclude population calibration claims. No probability recalibration was fitted to these outcomes.

TABLE XXIII

CONDITION B MODAL-CONFIDENCE BINS OVER 29 ALIGNED EVENTS PER CAMPAIGN (T1 EXCLUDED), EVALUATOR A AVERAGED VECTORS. BINS ARE LEFT-CLOSED/RIGHT-OPEN EXCEPT THE FINAL BIN, WHICH INCLUDES ONE.

Campaign	Confidence bin	Events	Mean confidence	Tie-aware accuracy
Llama-3.1-8B	0.00–0.25	0	—	—
Llama-3.1-8B	0.25–0.50	12	0.385	0.708
Llama-3.1-8B	0.50–0.75	11	0.620	0.818
Llama-3.1-8B	0.75–1.00	6	0.805	0.833
Mistral-7B	0.00–0.25	0	—	—
Mistral-7B	0.25–0.50	5	0.422	0.800
Mistral-7B	0.50–0.75	17	0.623	0.647
Mistral-7B	0.75–1.00	7	0.797	0.857
Qwen2.5-7B	0.00–0.25	0	—	—
Qwen2.5-7B	0.25–0.50	3	0.397	1.000
Qwen2.5-7B	0.50–0.75	20	0.615	0.800
Qwen2.5-7B	0.75–1.00	6	0.827	1.000
Qwen2.5-14B	0.00–0.25	0	—	—
Qwen2.5-14B	0.25–0.50	3	0.334	0.667
Qwen2.5-14B	0.50–0.75	19	0.617	0.947
Qwen2.5-14B	0.75–1.00	7	0.847	1.000

D. Llama-3.1-8B: full per-event records

TABLE XXIV

LLAMA-3.1-8B: ALL 30 STORED EVENT IDS; QWEN2.5-14B T1 IS A DIFFERENT QUESTION AND IS NOT A VALID CROSS-CAMPAIGN PAIR. BS A/B/C AND ACCURACY USE EVALUATOR A'S AVERAGED VECTOR. LAST COLUMN USES THE SINGLE EVALUATOR C READING OF CONDITION B. THESE ARE TWO DIFFERENT EXTRACTION BUDGETS. READ COUNT IS FOR B; STALE EXCLUSIONS IN OTHER CONDITIONS REMAIN APPLIED.

Event	BS A	BS B	BS C	C minus B	Acc. B	Reads B	Eval. C BS B
C1	0.5052	0.4747	0.4056	-0.0691	1.000	4	0.2150
C2	0.0039	0.3236	0.2314	-0.0923	1.000	4	0.2648
C3	1.0793	0.4336	1.0672	+0.6336	1.000	4	1.3056
C4	0.0112	0.0383	0.0183	-0.0200	1.000	4	0.0014
C5	0.4449	0.8472	1.0225	+0.1753	0.000	4	0.0450
C6	0.8247	0.7424	0.9772	+0.2348	0.000	4	0.5162
C7	0.3102	0.5322	0.3807	-0.1514	1.000	4	0.5918
C8	0.4601	0.7137	0.8016	+0.0878	0.500	4	0.7678
C9	0.2096	0.6284	0.8350	+0.2066	1.000	4	0.4826
C10	0.0036	0.0439	0.2966	+0.2527	1.000	4	0.0012
C11	1.0547	0.6092	0.5300	-0.0792	1.000	4	0.1000
C12	0.0006	0.1137	0.8450	+0.7313	1.000	4	0.0250
C13	0.2259	0.3115	0.5137	+0.2021	1.000	4	0.5906
C14	0.0600	0.7875	0.4590	-0.3285	0.000	4	0.1154
C15	0.0050	0.3105	0.6258	+0.3153	1.000	4	0.4806
S2	0.0450	0.0800	1.0878	+1.0078	1.000	4	0.0450
S3	0.7200	0.6903	0.9800	+0.2897	0.000	4	0.0648
S4	0.6050	0.1800	0.0528	-0.1272	1.000	4	0.0968
S5	0.4278	0.0800	0.2701	+0.1901	1.000	4	0.0648
S6	0.0528	1.4478	1.1562	-0.2916	0.000	4	1.0776
S7	0.2556	0.4050	0.5778	+0.1728	1.000	4	0.2450
S9	0.6328	1.2561	0.8128	-0.4433	0.000	4	1.2800
T1	0.0312	0.3403	0.3828	+0.0425	1.000	4	0.1568
T2	1.3203	0.4141	0.1953	-0.2187	1.000	4	0.8450
T3	0.4587	0.1621	0.3116	+0.1495	1.000	4	0.0776
T4	0.0350	0.1866	0.1095	-0.0771	1.000	4	0.0182
T5	0.1300	0.1711	0.0231	-0.1480	1.000	4	0.0200
T6	0.6128	0.4267	0.5834	+0.1567	1.000	4	0.3050
T8	0.4222	0.4016	0.3042	-0.0974	1.000	4	0.2718
T9	0.0013	0.0903	0.0761	-0.0143	1.000	4	0.0800

TABLE XXV

LLAMA-3.1-8B: PRIMARY-FILE BRIER AND SURVIVING EVIDENCE SIZE. PRIMARY-FILE READINGS ARE INSTRUMENT-MIXED FOR RECOVERED LLAMA AND QWEN2.5-14B UNITS AND MUST NOT ALL BE CALLED EVALUATOR J. LAST TWO COLUMNS DESCRIBE A'S AVERAGED CONDITION B VECTOR. CHARACTER COUNT DESCRIBES SCORED EVIDENCE, NOT ALL GENERATED DISCUSSION.

Event	Primary BS B	Primary BS C	Chars B	Chars C	A max prob.	A uniform
C1	1.0550	0.5150	19900	19158	0.463	no
C2	0.9200	1.3400	18157	19827	0.500	no
C3	0.7500	0.7500	19818	19414	0.450	no
C4	0.5000	0.0342	18638	19546	0.833	no
C5	1.1450	1.0800	19804	19642	0.300	no
C6	1.4150	1.1950	20303	20221	0.380	no
C7	0.7000	0.8333	20150	17694	0.345	no
C8	0.8000	0.8200	20017	19790	0.250	no
C9	0.6534	0.2900	19925	20518	0.275	no
C10	0.0036	0.2834	19865	20454	0.820	no
C11	1.3400	0.7550	20907	19593	0.305	no
C12	0.1150	0.1034	20372	19242	0.705	no
C13	1.3880	1.2500	19572	18825	0.497	no
C14	1.0295	0.6100	20745	19984	0.475	no
C15	0.9100	0.5700	20489	20156	0.507	no
S2	0.5000	0.5000	18381	18815	0.800	no
S3	0.5000	0.5000	19816	20046	0.588	no
S4	0.5000	0.5000	19135	20035	0.700	no
S5	0.5000	0.5000	18958	19075	0.800	no
S6	0.7500	0.7500	20329	19126	0.712	no
S7	0.9800	0.7200	19435	19675	0.550	no
S9	0.9800	0.9800	19485	19646	0.792	no
T1	0.1800	0.3200	19782	20594	0.588	no
T2	0.0450	0.0800	19383	19288	0.545	no
T3	0.2850	0.2850	18855	20125	0.680	no
T4	1.2000	0.1150	19493	19065	0.620	no
T5	0.0450	0.0450	19761	20331	0.708	no
T6	0.7700	0.9000	20228	19100	0.435	no
T8	0.5100	0.2538	19691	19805	0.450	no
T9	0.0800	0.0200	19466	19177	0.787	no

E. Mistral-7B: full per-event records

TABLE XXVI

MISTRAL-7B: ALL 30 STORED EVENT IDS; QWEN2.5-14B T1 IS A DIFFERENT QUESTION AND IS NOT A VALID CROSS-CAMPAIGN PAIR. BS A/B/C AND ACCURACY USE EVALUATOR A'S AVERAGED VECTOR. LAST COLUMN USES THE SINGLE EVALUATOR C READING OF CONDITION B. THESE ARE TWO DIFFERENT EXTRACTION BUDGETS. READ COUNT IS FOR B; STALE EXCLUSIONS IN OTHER CONDITIONS REMAIN APPLIED.

Event	BS A	BS B	BS C	C minus B	Acc. B	Reads B	Eval. C BS B
C1	0.3890	0.4142	0.6028	+0.1886	1.000	4	0.3294
C2	0.0038	0.8076	0.1400	-0.6675	0.000	4	0.1018
C3	1.0268	0.3064	0.4531	+0.1467	1.000	4	0.1146
C4	0.0009	0.7734	1.1959	+0.4225	0.000	4	1.4494
C5	0.4859	0.3712	0.5791	+0.2078	1.000	4	0.4416
C6	0.5458	0.2494	0.9991	+0.7497	1.000	4	0.2016
C7	0.2544	1.1238	1.5954	+0.4716	0.000	4	0.8152
C8	0.3993	0.6166	0.7384	+0.1219	1.000	4	1.0588
C9	0.2809	0.3461	0.8529	+0.5068	1.000	4	0.1668
C10	0.0026	0.1048	0.3650	+0.2602	1.000	4	0.0142
C11	0.4361	0.1039	1.0134	+0.9095	1.000	4	0.0256
C12	0.0068	1.4302	1.6506	+0.2204	0.000	4	1.1942
C13	0.1852	0.0647	0.6622	+0.5975	1.000	4	0.0410
C14	0.0078	0.4837	0.1453	-0.3384	1.000	4	0.2106
C15	0.0066	0.1320	0.1887	+0.0567	1.000	4	0.0514
S2	0.0465	0.0946	0.2701	+0.1755	1.000	4	0.2048
S3	0.6962	0.4851	0.4278	-0.0573	1.000	4	0.3528
S4	0.3613	0.8778	0.2048	-0.6730	0.000	4	1.1250
S5	0.4371	0.0378	0.2112	+0.1734	1.000	4	0.0288
S6	0.0608	1.3105	0.6489	-0.6617	0.000	4	0.0142
S7	0.2556	0.1128	0.3612	+0.2484	1.000	4	0.0968
S9	0.3828	0.2112	0.3003	+0.0891	1.000	4	0.2048
T1	0.0021	0.8450	0.3828	-0.4622	0.000	4	1.0368
T2	1.4028	0.0528	0.1250	+0.0722	1.000	4	0.0128
T3	0.2009	1.0541	0.2738	-0.7803	0.000	4	0.6200
T4	0.0353	0.2734	0.0849	-0.1885	1.000	4	0.0416
T5	0.0066	0.1800	0.0968	-0.0832	1.000	4	0.0512
T6	0.3901	1.3159	0.2755	-1.0404	0.000	4	1.1360
T8	0.1809	0.1624	0.1169	-0.0455	1.000	4	0.2370
T9	0.0078	0.1128	0.1012	-0.0116	1.000	4	0.0648

TABLE XXVII

MISTRAL-7B: PRIMARY-FILE BRIER AND SURVIVING EVIDENCE SIZE. PRIMARY-FILE READINGS ARE INSTRUMENT-MIXED FOR RECOVERED LLAMA AND QWEN2.5-14B UNITS AND MUST NOT ALL BE CALLED EVALUATOR J. LAST TWO COLUMNS DESCRIBE A'S AVERAGED CONDITION B VECTOR. CHARACTER COUNT DESCRIBES SCORED EVIDENCE, NOT ALL GENERATED DISCUSSION.

Event	Primary BS B	Primary BS C	Chars B	Chars C	A max prob.	A uniform
C1	0.6667	0.5600	14719	12814	0.500	no
C2	1.1338	0.6200	12089	12953	0.575	no
C3	0.7000	1.1350	12984	12907	0.537	no
C4	0.1368	0.5000	12007	11974	0.575	no
C5	0.8000	0.8000	12996	13064	0.463	no
C6	0.6650	0.8000	12042	12771	0.575	no
C7	0.8000	0.7300	11566	10192	0.575	no
C8	0.7500	0.8000	12178	13942	0.312	no
C9	0.5950	0.8100	13517	11572	0.450	no
C10	0.8000	0.6950	14241	12194	0.712	no
C11	0.6250	0.8000	11728	13236	0.713	no
C12	0.5926	1.2000	11711	10930	0.800	no
C13	0.0480	0.6000	10203	10814	0.775	no
C14	0.7250	0.7150	12785	11634	0.438	no
C15	0.2083	0.3000	11070	11380	0.675	no
S2	0.5202	0.5000	11148	13011	0.782	no
S3	0.3200	0.5000	13222	13906	0.508	no
S4	0.7200	0.5000	11794	10446	0.662	no
S5	0.3200	0.3200	12467	13318	0.863	no
S6	0.7500	0.7500	15557	15071	0.730	no
S7	0.1800	1.1250	12777	14013	0.762	no
S9	0.0800	0.5000	13881	14486	0.675	no
T1	0.5000	0.5000	15348	15253	0.650	no
T2	0.0200	0.5000	13129	14496	0.838	no
T3	0.7800	0.4650	11433	12606	0.450	no
T4	0.5300	0.4600	11629	11978	0.537	no
T5	0.5000	0.1800	13728	12074	0.700	no
T6	0.7500	0.9000	13692	12750	0.688	no
T8	0.5400	0.8000	13081	13041	0.660	no
T9	0.1800	0.3200	11980	13388	0.762	no

F. Qwen2.5-7B: full per-event records

TABLE XXVIII

QWEN2.5-7B: ALL 30 STORED EVENT IDS; QWEN2.5-14B T1 IS A DIFFERENT QUESTION AND IS NOT A VALID CROSS-CAMPAIGN PAIR. BS A/B/C AND ACCURACY USE EVALUATOR A'S AVERAGED VECTOR. LAST COLUMN USES THE SINGLE EVALUATOR C READING OF CONDITION B. THESE ARE TWO DIFFERENT EXTRACTION BUDGETS. READ COUNT IS FOR B; STALE EXCLUSIONS IN OTHER CONDITIONS REMAIN APPLIED.

Event	BS A	BS B	BS C	C minus B	Acc. B	Reads B	Eval. C BS B
C1	0.3941	0.3726	0.5329	+0.1602	1.000	4	0.2616
C2	0.0026	0.0249	0.1050	+0.0801	1.000	4	0.0424
C3	1.0610	0.4114	0.4147	+0.0033	1.000	4	0.0050
C4	0.0004	0.0294	0.1056	+0.0761	1.000	4	0.0014
C5	0.4672	0.8237	0.4366	-0.3870	0.000	4	0.3086
C6	0.5528	0.5662	0.7726	+0.2064	1.000	4	0.2406
C7	0.2918	0.2041	0.7233	+0.5193	1.000	4	0.0610
C8	0.4054	0.1759	0.8109	+0.6350	1.000	4	0.5470
C9	0.1937	0.2451	0.4163	+0.1712	1.000	4	0.1710
C10	0.0032	0.1213	0.1585	+0.0372	1.000	4	0.0422
C11	0.5402	0.1456	0.6231	+0.4775	1.000	4	0.1306
C12	0.0043	0.0710	0.1237	+0.0527	1.000	4	0.0406
C13	0.2036	0.1658	0.5545	+0.3887	1.000	4	0.0286
C14	0.0147	0.1520	0.1651	+0.0132	1.000	4	0.1496
C15	0.0117	0.1440	0.2932	+0.1493	1.000	4	0.1668
S2	0.0450	0.2926	0.2450	-0.0476	1.000	4	0.0968
S3	0.6962	0.9941	1.1858	+0.1917	0.000	4	0.6728
S4	0.3613	0.3655	0.5995	+0.2340	1.000	4	0.0968
S5	0.4656	0.3321	0.7140	+0.3819	1.000	4	0.0968
S6	0.0642	0.2319	0.3120	+0.0801	1.000	4	0.1208
S7	0.2850	0.3200	0.5151	+0.1951	1.000	4	0.2048
S9	0.5778	0.7812	1.2403	+0.4591	0.000	4	0.3200
T1	0.0066	0.2628	0.2628	+0.0000	1.000	4	0.2048
T2	1.1628	0.0253	0.1250	+0.0997	1.000	4	0.0200
T3	0.2034	0.1503	0.1372	-0.0131	1.000	4	0.1400
T4	0.0193	0.0791	0.0872	+0.0081	1.000	4	0.0020
T5	0.0153	0.6328	0.8128	+0.1800	0.000	4	0.1568
T6	0.5392	0.4741	0.6407	+0.1667	1.000	4	0.2900
T8	0.3068	0.2948	0.1553	-0.1395	1.000	4	0.2424
T9	0.0050	0.0481	0.0685	+0.0204	1.000	4	0.0800

TABLE XXIX

QWEN2.5-7B: PRIMARY-FILE BRIER AND SURVIVING EVIDENCE SIZE. PRIMARY-FILE READINGS ARE INSTRUMENT-MIXED FOR RECOVERED LLAMA AND QWEN2.5-14B UNITS AND MUST NOT ALL BE CALLED EVALUATOR J. LAST TWO COLUMNS DESCRIBE A'S AVERAGED CONDITION B VECTOR. CHARACTER COUNT DESCRIBES SCORED EVIDENCE, NOT ALL GENERATED DISCUSSION.

Event	Primary BS B	Primary BS C	Chars B	Chars C	A max prob.	A uniform
C1	0.4650	0.3350	7918	6884	0.542	no
C2	0.1150	0.6400	6821	6679	0.860	no
C3	0.5850	0.9000	7544	6911	0.450	no
C4	0.1250	0.1650	7309	7115	0.855	no
C5	0.1400	0.6531	7366	6738	0.542	no
C6	1.0450	1.0550	7454	8362	0.328	no
C7	0.5300	0.7950	7455	6507	0.600	no
C8	1.2000	0.6200	7165	6546	0.637	no
C9	0.6032	0.5750	8184	6876	0.550	no
C10	0.6400	0.4800	7072	7031	0.705	no
C11	0.5500	1.0550	7196	6646	0.662	no
C12	0.1150	0.1338	7872	6758	0.762	no
C13	0.6100	0.4438	6785	7027	0.637	no
C14	0.2250	0.5000	7147	7346	0.667	no
C15	0.6750	0.6350	7418	6934	0.667	no
S2	0.1800	0.3200	6981	6701	0.617	no
S3	0.8450	0.9800	7990	7011	0.705	no
S4	0.3200	0.1800	7452	8179	0.573	no
S5	0.3200	0.3200	7525	6793	0.593	no
S6	0.4150	0.5000	8071	6690	0.588	no
S7	0.1250	0.3200	7229	6722	0.600	no
S9	0.3200	0.9800	7404	6939	0.625	no
T1	0.9800	0.3200	7300	7414	0.637	no
T2	0.0050	0.7200	7208	6790	0.887	no
T3	0.1950	0.4550	6914	6681	0.688	no
T4	0.6200	0.2150	7038	7075	0.750	no
T5	0.6050	0.3200	7473	6946	0.562	no
T6	1.0150	0.7850	6860	6591	0.412	no
T8	0.5550	0.5550	7805	6857	0.542	no
T9	0.3200	0.0800	6969	6584	0.845	no

G. Qwen2.5-14B: full per-event records

TABLE XXX

QWEN2.5-14B: ALL 30 STORED EVENT IDS; QWEN2.5-14B T1 IS A DIFFERENT QUESTION AND IS NOT A VALID CROSS-CAMPAIGN PAIR. BS A/B/C AND ACCURACY USE EVALUATOR A'S AVERAGED VECTOR. LAST COLUMN USES THE SINGLE EVALUATOR C READING OF CONDITION B. THESE ARE TWO DIFFERENT EXTRACTION BUDGETS. READ COUNT IS FOR B; STALE EXCLUSIONS IN OTHER CONDITIONS REMAIN APPLIED.

Event	BS A	BS B	BS C	C minus B	Acc. B	Reads B	Eval. C BS B
C1	0.3844	0.3106	0.4373	+0.1267	1.000	3	0.2558
C2	0.0020	0.0286	0.1232	+0.0947	1.000	3	0.0186
C3	1.2346	0.3389	0.4594	+0.1206	1.000	3	0.2948
C4	0.0008	0.0062	0.0541	+0.0479	1.000	3	0.0023
C5	0.4852	0.2544	0.2373	-0.0172	1.000	3	0.2336
C6	0.2611	1.1463	1.0933	-0.0530	0.000	3	1.0086
C7	0.2732	0.1155	0.3886	+0.2731	1.000	3	0.0402
C8	0.4033	0.1450	0.5483	+0.4034	1.000	3	0.4402
C9	0.1940	0.1687	0.3670	+0.1984	1.000	3	0.2382
C10	0.0040	0.0538	0.1312	+0.0774	1.000	3	0.0272
C11	0.9659	0.2978	0.2239	-0.0739	1.000	3	0.0618
C12	0.0045	0.0425	0.0694	+0.0269	1.000	3	0.0125
C13	0.2192	0.0278	0.6811	+0.6533	1.000	3	0.0308
C14	0.0039	0.1533	0.2183	+0.0650	1.000	3	0.0696
C15	0.0149	0.1854	0.4983	+0.3129	1.000	3	0.1338
S2	0.0450	0.1352	0.4294	+0.2942	1.000	3	0.0648
S3	0.7120	0.4544	0.6806	+0.2261	1.000	3	0.7200
S4	0.4672	0.2544	0.1881	-0.0663	1.000	3	0.3528
S5	0.4672	0.5760	0.4171	-0.1589	0.000	3	0.1568
S6	0.0671	0.1014	0.1550	+0.0536	1.000	3	0.0440
S7	0.3200	0.1284	0.0450	-0.0834	1.000	3	0.1568
S9	0.3472	0.4868	0.6498	+0.1630	1.000	3	0.2888
T1	0.0027	0.2006	0.5202	+0.3196	1.000	3	0.0512
T2	1.2272	0.0272	0.4050	+0.3778	1.000	3	0.8450
T3	0.2672	0.6534	0.4651	-0.1883	1.000	3	0.4650
T4	0.0204	0.2238	0.0262	-0.1976	1.000	3	0.0246
T5	0.0118	0.3641	0.3472	-0.0169	1.000	3	0.0968
T6	0.4114	0.6976	0.7239	+0.0263	1.000	3	0.3926
T8	0.2729	0.1403	0.4508	+0.3105	1.000	3	0.2824
T9	0.0128	0.0854	0.1840	+0.0986	1.000	3	0.1800

TABLE XXXI

QWEN2.5-14B: PRIMARY-FILE BRIER AND SURVIVING EVIDENCE SIZE. PRIMARY-FILE READINGS ARE INSTRUMENT-MIXED FOR RECOVERED LLAMA AND QWEN2.5-14B UNITS AND MUST NOT ALL BE CALLED EVALUATOR J. LAST TWO COLUMNS DESCRIBE A'S AVERAGED CONDITION B VECTOR. CHARACTER COUNT DESCRIBES SCORED EVIDENCE, NOT ALL GENERATED DISCUSSION.

Event	Primary BS B	Primary BS C	Chars B	Chars C	A max prob.	A uniform
C1	0.1400	0.5600	6908	6581	0.583	no
C2	0.2338	0.1700	6982	6698	0.850	no
C3	0.3450	0.6850	7132	6783	0.517	no
C4	0.1350	0.5000	6952	6611	0.933	no
C5	0.1338	0.5150	7012	6704	0.550	no
C6	1.0600	1.1200	9861	9628	0.387	no
C7	0.6000	0.4990	6908	6582	0.700	no
C8	0.6400	0.7000	6853	6550	0.673	no
C9	0.6827	0.5596	7032	6699	0.617	no
C10	0.2494	0.6850	6875	6607	0.800	no
C11	0.3750	1.1950	6961	6663	0.517	no
C12	0.1150	0.6350	6958	6569	0.817	no
C13	0.0270	0.8950	6661	6441	0.850	no
C14	0.5000	0.5750	7087	6820	0.667	no
C15	0.4000	0.7150	6962	6636	0.617	no
S2	0.1800	0.3528	6911	6683	0.740	no
S3	0.6050	0.4050	6951	6816	0.523	no
S4	0.1800	0.1250	6931	6722	0.643	no
S5	0.6050	0.6050	6822	6624	0.537	no
S6	0.5650	0.0918	6983	6859	0.733	no
S7	0.1250	0.1250	6940	6666	0.747	no
S9	0.1800	0.3200	6997	6719	0.507	no
T1	0.1250	0.3200	6958	6683	0.683	no
T2	0.1800	0.7200	6959	6667	0.883	no
T3	0.5400	0.5400	10273	10093	0.340	no
T4	0.0136	0.0032	10711	12558	0.600	no
T5	0.3200	0.3200	10770	11336	0.573	no
T6	0.7500	0.7500	9593	8633	0.277	no
T8	0.0888	0.6400	8829	9431	0.683	no
T9	0.1250	0.6050	6849	6509	0.793	no

APPENDIX J REPRODUCTION AND ARTIFACT CONTRACT

The accompanying archive contains the benchmark definition in `01_benchmark`, scored records in `02_campaigns`, available traces in `03_traces`, existing follow-up observations in `04_experiments`, revised analysis in `05_analysis`, historical execution code in `07_pipeline` and superseded artifacts in `09_audit`. From the archive root:

```
python 05_analysis/scripts/paper_analysis.py -data-root . -output-dir 08_paper
```

The script reads frozen observations and generates `paper_results.json`, `paper_tables.tex` and `fig_paper_audit.png`, without network calls. The extended tables are generated separately by:

```
python 05_analysis/scripts/paper_full_analysis.py -data-root . -output-dir 08_paper
```

This produces `paper_full_results.json` and `paper_full_tables.tex`; it checks question alignment and retains the T1 mismatch rather than silently averaging different questions. The JSON records settings and SHA-256 input hashes; tables consume generated values directly. A numerical validation suite checks estimands and data-contract failure cases.

The paired comparison validates evidence/question/options equality, not complete historical request equality. Four stale rereads are excluded by file-unit keys. Forecast metrics are recomputed from vectors rather than inherited fields. Historical scripts and reports are retained for provenance; they are not authoritative for current paper tables.

APPENDIX K CORRECTION RECORD

The prior manuscript is retained under `09_audit/manuscript_20260826.tex` and is superseded. This revision replaces a demonstrated-mechanism claim with a fixed-evidence configuration observation; replaces contamination-clean language with post-release subset; withdraws the independent-unit pooled accuracy test; uses event-cluster intervals; adds evaluator, aggregation, input and dependency sensitivities; corrects the B/C description; retains uniform vectors in scoring; and withdraws unvalidated convergence, size-threshold and optimal-configuration claims while restoring their underlying design and engineering context in full. The extended question audit adds the surviving Qwen2.5-14B T1 mismatch. The long-form source, rubric definitions, scaling matrices, run-integrity discussion and original figure inventory are preserved with their evidential status stated, rather than replaced by a short case study. No new model observations were collected. The defects remain available to audit and remain explicit limits on inference.

APPENDIX L

HISTORICAL FIGURE REGISTER WITH CORRECTED INTERPRETATIONS

All 30 distinct figures used in the long August manuscript are retained in this register so that its visual research record is not lost. These are documentary reproductions, not additional current results. Their embedded titles, p-values, category names and mechanistic annotations belong to the earlier analysis. The explicit note beside each plate states the corrected interpretation; current numerical authority remains the generated tables and the main audit figure. Retaining an old image does not endorse its superseded claim. The bitmap files are unchanged in the historical figure directory.

Pipeline architecture

Historical artifact 1/30; not a current result. The schematic records the intended pipeline and role separation. It does not certify execution integrity, per-model reference calls, length-matched null inputs, or validated behavioral scores.

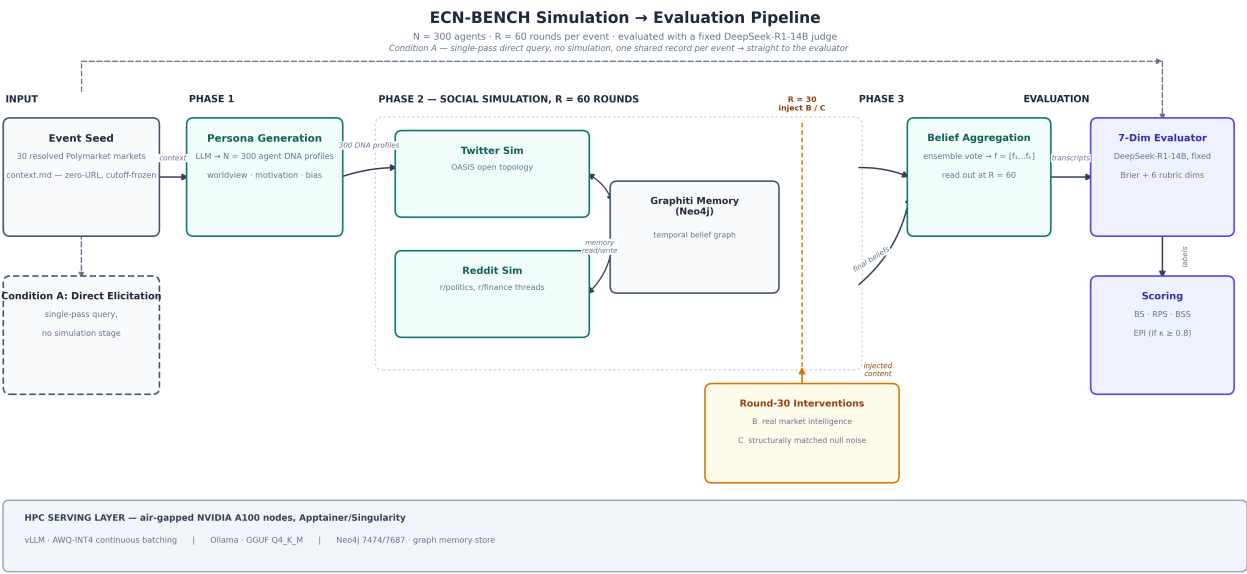


Fig. 3. Documentary reproduction: pipeline architecture. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig8_architecture.png`.

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Historical artifact 2/30; not a current result. A-to-B lift uses a shared reference and historically mixed evaluator/recovery states. Color clusters are descriptive, not evidence of a within-model deliberation benefit. Regenerated per-event tables supersede its numerical interpretation.

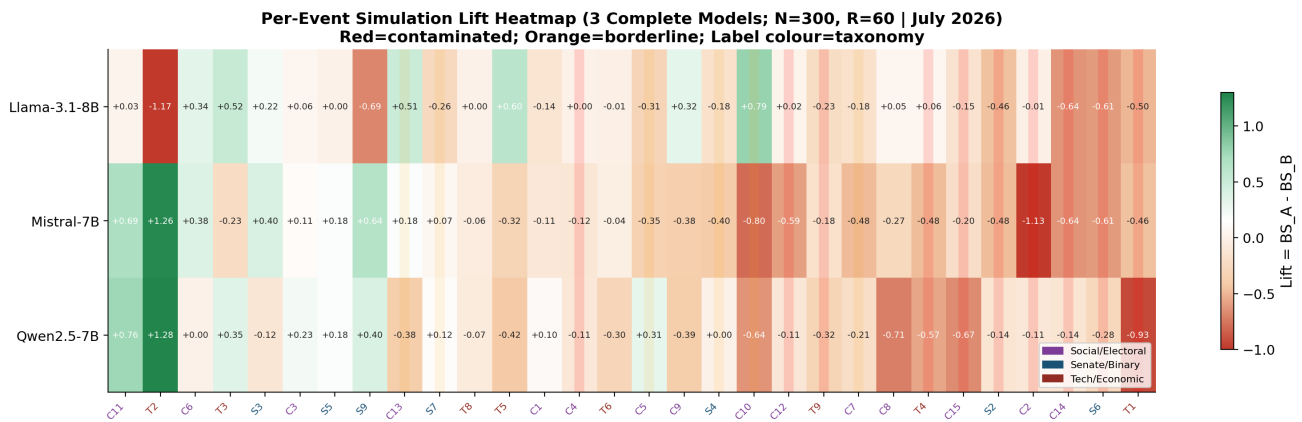


Fig. 4. Documentary reproduction: per-event reference contrasts. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig1_lift_heatmap.png`.

Original campaign summary

Historical artifact 3/30; not a current result. This is a historical plot of the earlier scoring pass. Its intervals and campaign ordering are not the current aligned analysis; model, serving and evaluator differences prevent an architectural ranking.

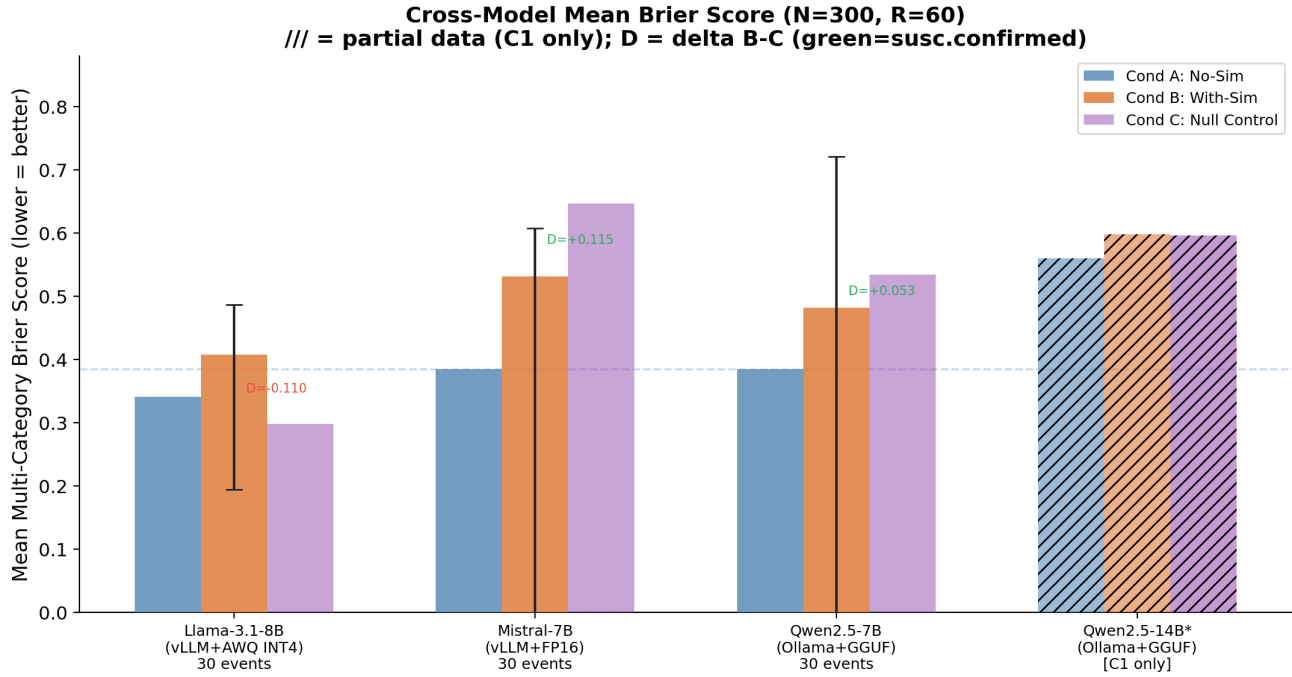


Fig. 5. Documentary reproduction: original campaign summary. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig2_cross_model_brier.png`.

Original B/C margins

Historical artifact 4/30; not a current result. The historical bars concern a whole-pipeline input contrast, not rational updating. Length matching and topic alignment were not achieved. Current event-cluster intervals are in the main sensitivity table.

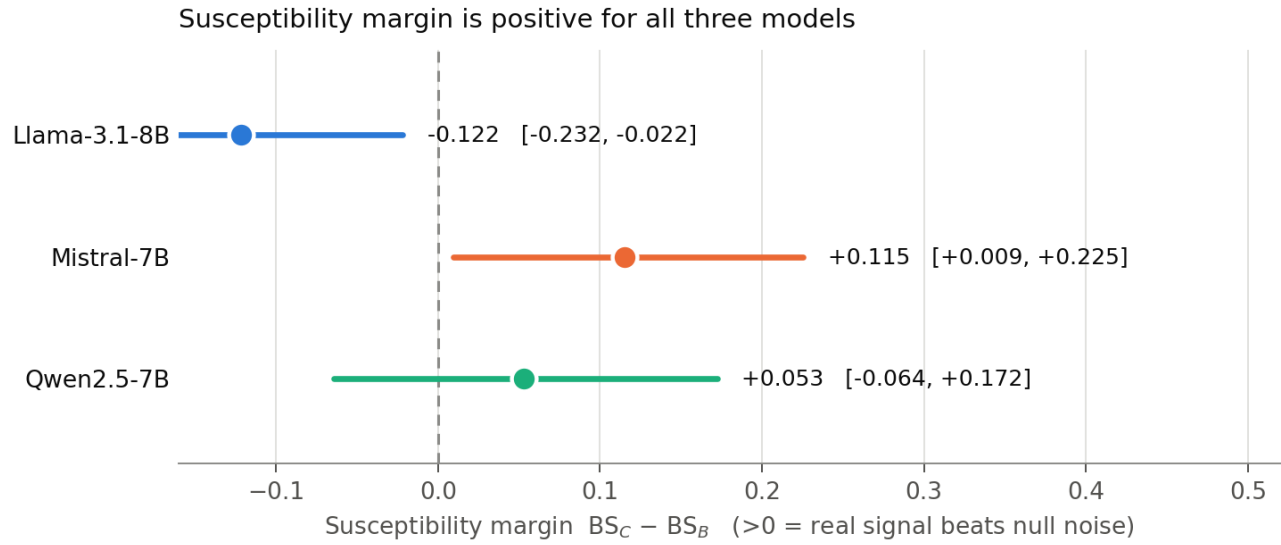


Fig. 6. Documentary reproduction: original b/c margins. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figB_susceptibility.png`.

Prefix-group comparison

Historical artifact 5/30; not a current result. C/S/T are mixed record categories, not a validated taxonomy. Earlier generalist/specialist interpretations are withdrawn. The regenerated prefix table uses the 29 question-aligned events and current A aggregation.

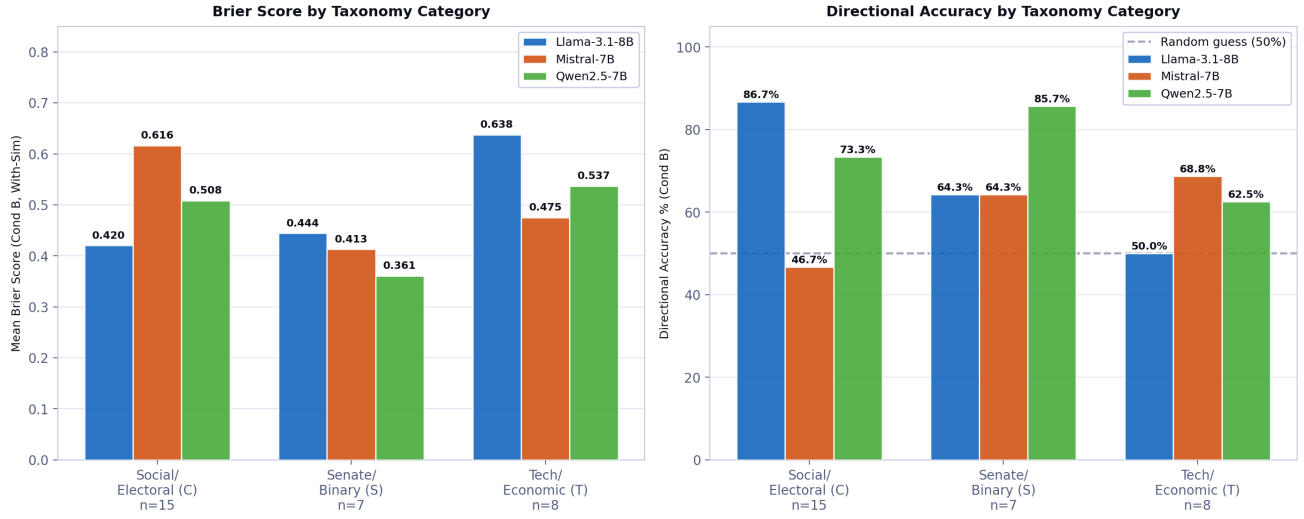


Fig. 7. Documentary reproduction: prefix-group comparison. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig3_taxonomy_breakdown.png`.

Original calibration rendering

Historical artifact 6/30; not a current result. This rendering is retained as an audit object, not evidence of systematic under-confidence or perfect calibration. Current confidence bins use modal probability, tie-aware accuracy and explicit occupancies.

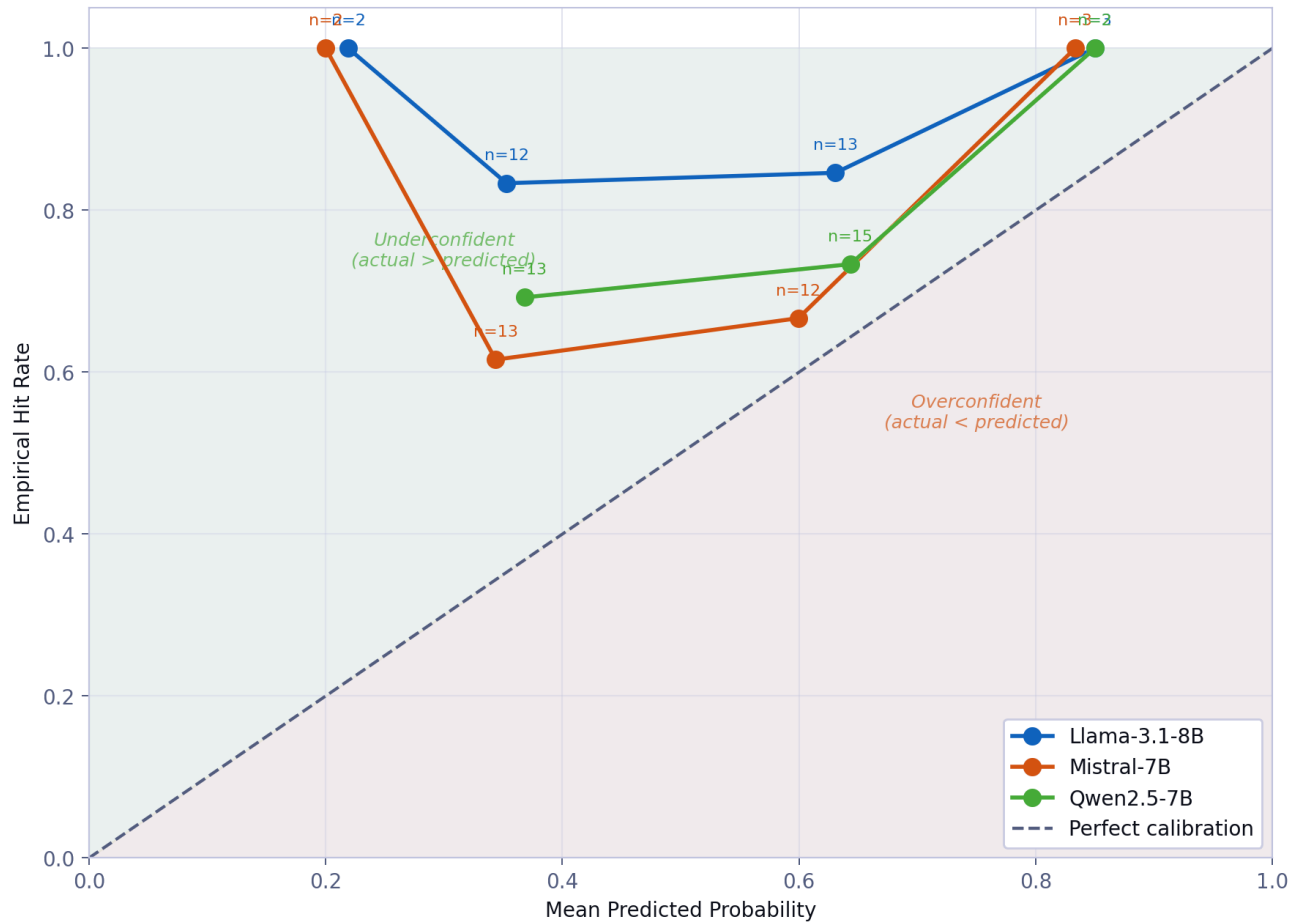


Fig. 8. Documentary reproduction: original calibration rendering. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig4_calibration_overlay.png`.

Near-uniformity and concentration

Historical artifact 7/30; not a current result. Uniform forecasts are valid probability vectors and remain in scoring. Reduced concentration does not necessarily worsen Brier. The configuration-matched 132-unit diagnostic supersedes pooled counts mixing recovery states.

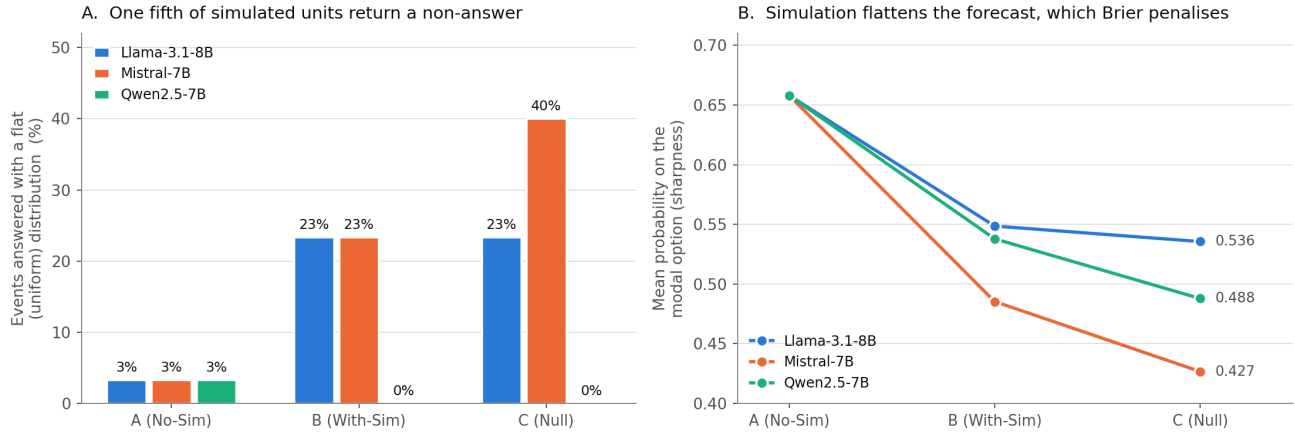


Fig. 9. Documentary reproduction: near-uniformity and concentration. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figE_degenerate_forecasts.png`.

Historical evaluator replication

Historical artifact 8/30; not a current result. The fixed-evidence comparison establishes sensitivity to extraction configuration. Neither non-uniformity nor JSON formatting proves fidelity; original and recovered units cannot all be counted as J/A pairs.

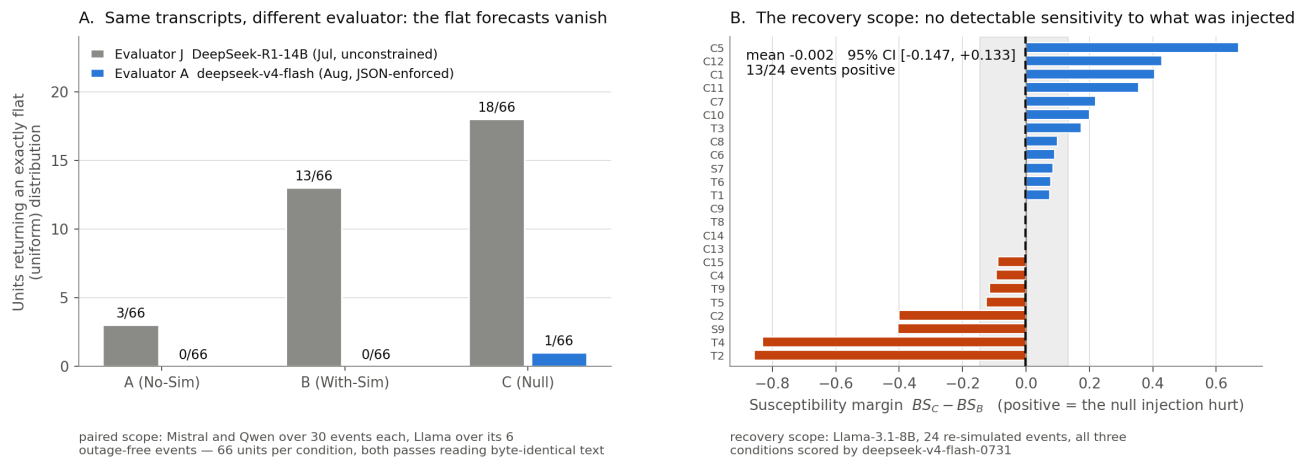


Fig. 10. Documentary reproduction: historical evaluator replication. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figR1_evaluator_replication.png`.

Historical repeated-read variability

Historical artifact 9/30; not a current result. Repeated requests are not independent simulations. Exact agreement and Brier range depend on available read counts. They do not define a universal noise floor or establish a minimum of three reads.

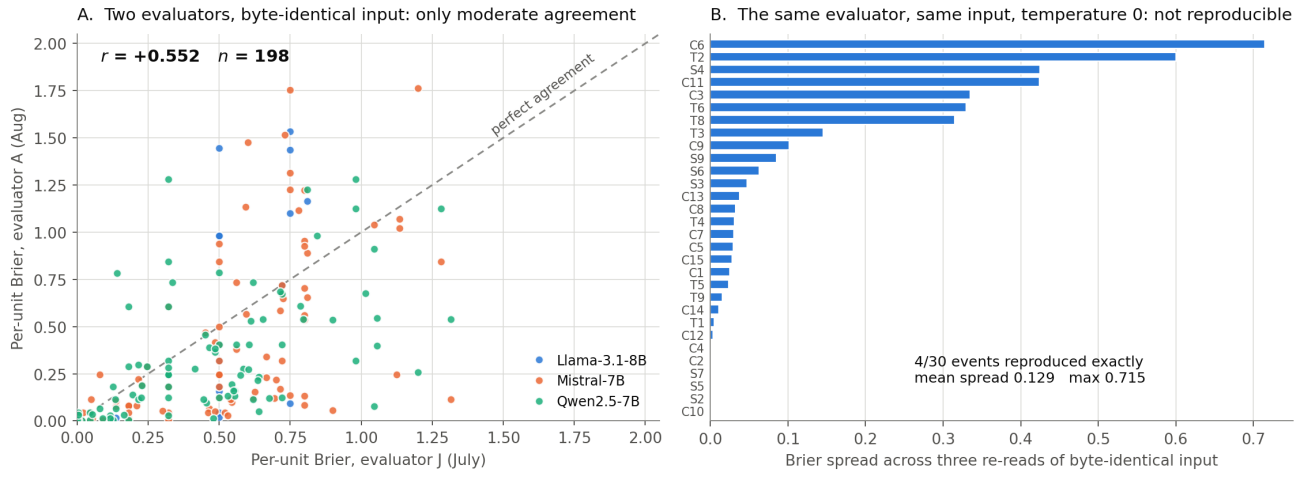


Fig. 11. Documentary reproduction: historical repeated-read variability. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figR2_instrument_noise.png`.

Evidence-length association

Historical artifact 10/30; not a current result. A length association is not a causal dose-response. Transcript content, campaign identity and serving differ. The proxy truncation grid did not isolate the mechanism of the historical J deployment.

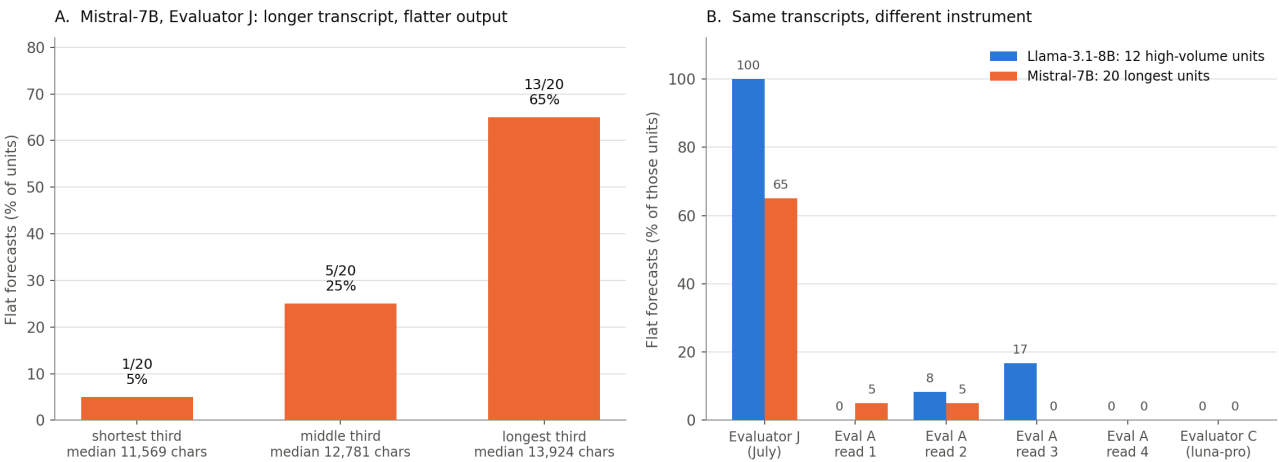


Fig. 12. Documentary reproduction: evidence-length association. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figI_volume_mechanism.png`.

Cross-evaluator transfer probe

Historical artifact 11/30; not a current result. Parsed-response denominators differ and missing responses are not assumed random. Prompt and serving differences preclude a clean model-size comparison. The former overlap significance claim is not retained as confirmatory inference.

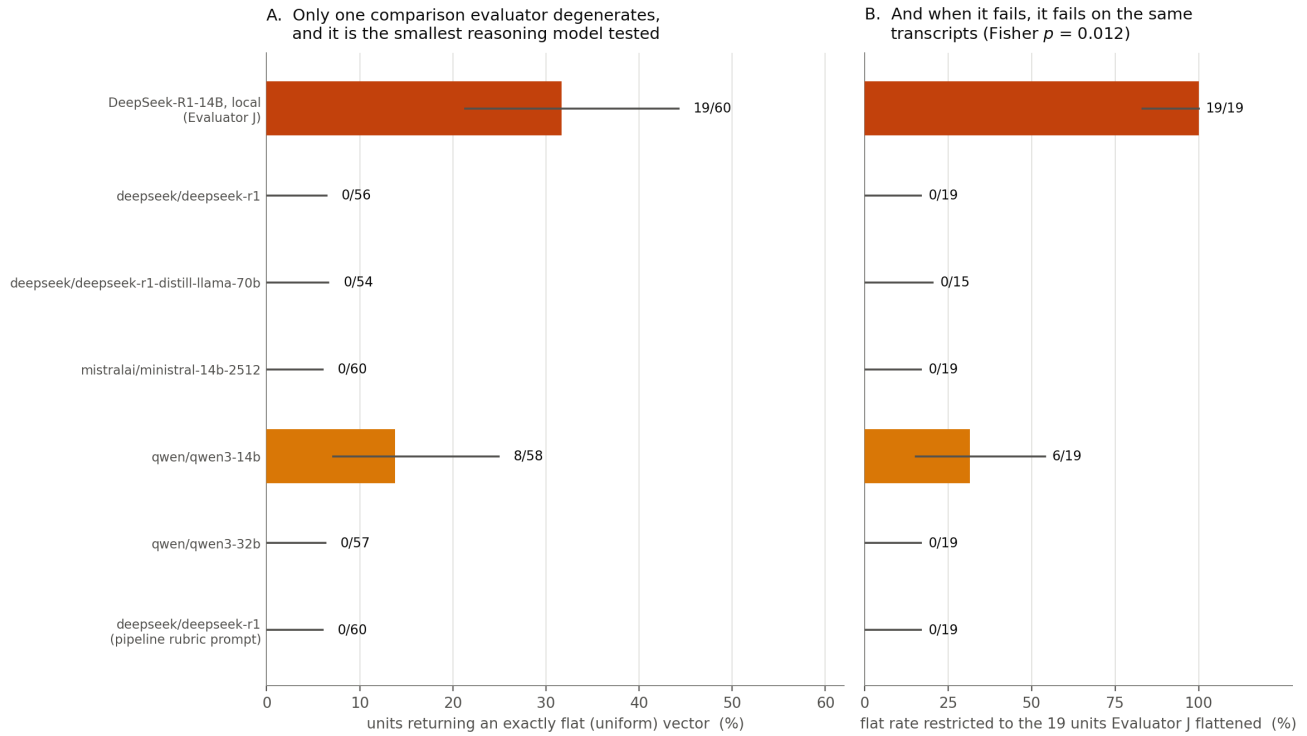


Fig. 13. Documentary reproduction: cross-evaluator transfer probe. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figK_degeneracy_transfer.png`.

Requested enforcement and length grid

Historical artifact 12/30; not a current result. The current reanalysis uses 238 complete off/on pairs clustered by event. The old pooled significance annotations do not establish equivalence or an absence of constraint effects. Truncation changes selected content as well as length.

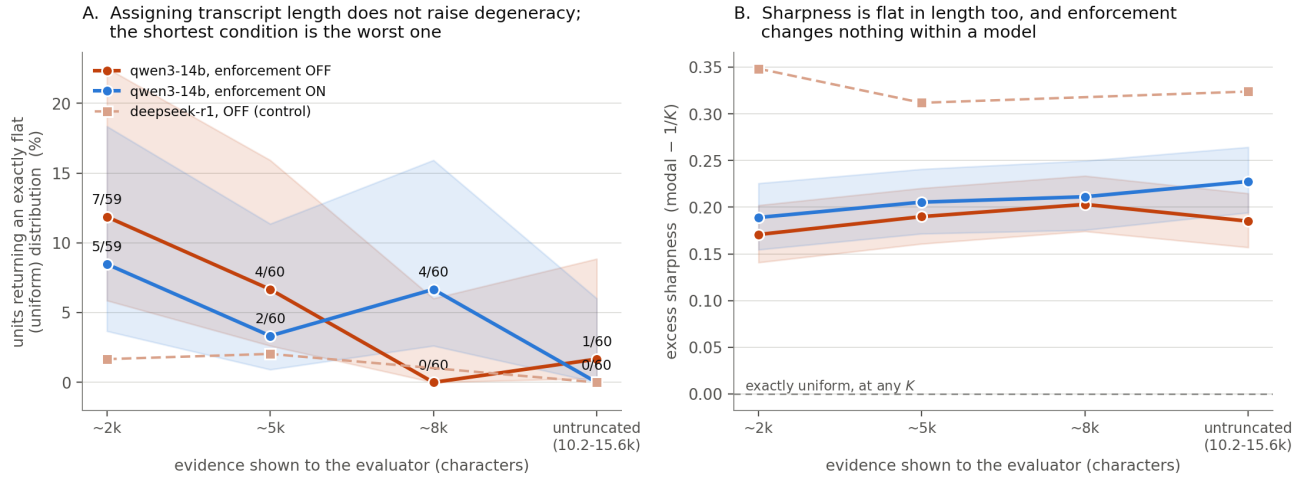


Fig. 14. Documentary reproduction: requested enforcement and length grid. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figL_truncation.png`.

Earlier forecasting summary

Historical artifact 13/30; not a current result. The extremely small pooled accuracy p-value is withdrawn because event repetitions are dependent and fractional tie credit was treated as an integer count. The post-release subset is not leakage-free. Current results and uncertainty replace those annotations.

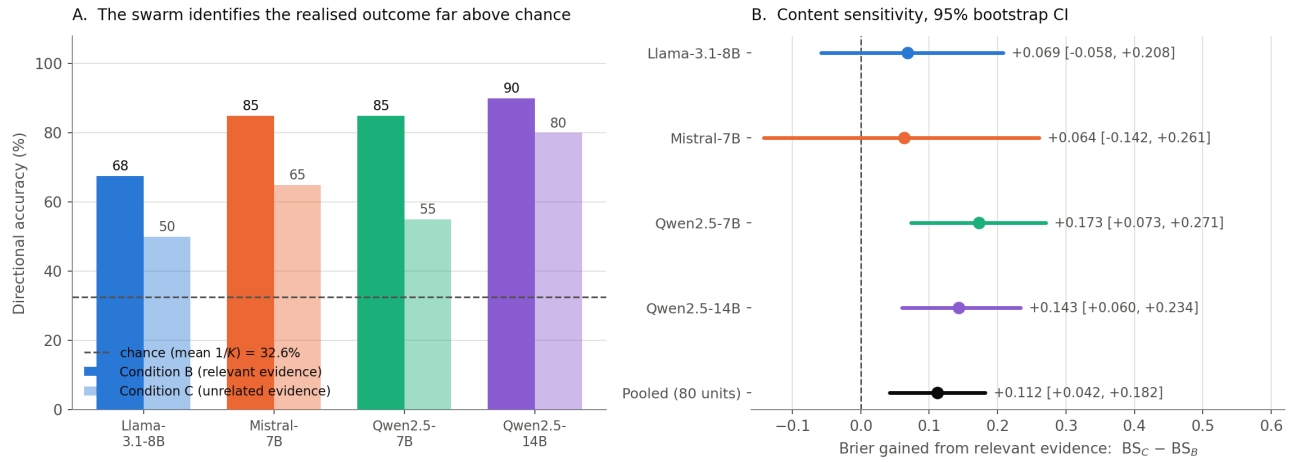


Fig. 15. Documentary reproduction: earlier forecasting summary. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figJ_predictive_signal.png`.

Single-agent comparison

Historical artifact 14/30; not a current result. The baseline received dossier and injection directly; no added retrieval pipeline was run. Only two arms nominally match campaign models, serving is unbalanced, and three temperatures/calls are not one fixed-policy replicate. No equivalence or swarm advantage is established.

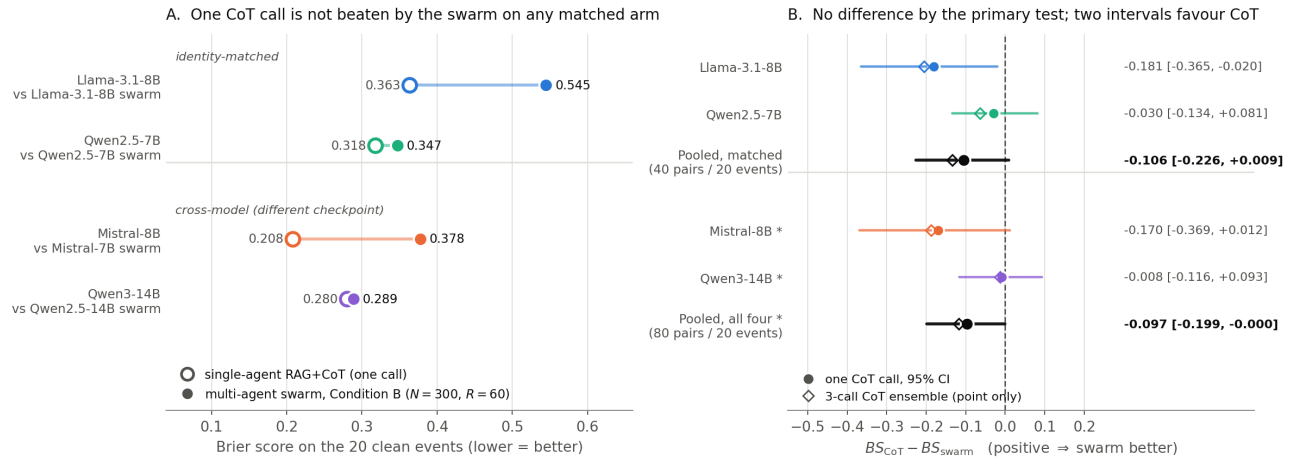


Fig. 16. Documentary reproduction: single-agent comparison. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figM_cot_vs_swarm.png`.

Telemetry integrity diagnostic

Historical artifact 15/30; not a current result. Constant initialized values can satisfy a non-strict monotonicity check. This is an integrity warning, not proof of belief convergence. Counts describe the historical telemetry version, not every recovered execution.

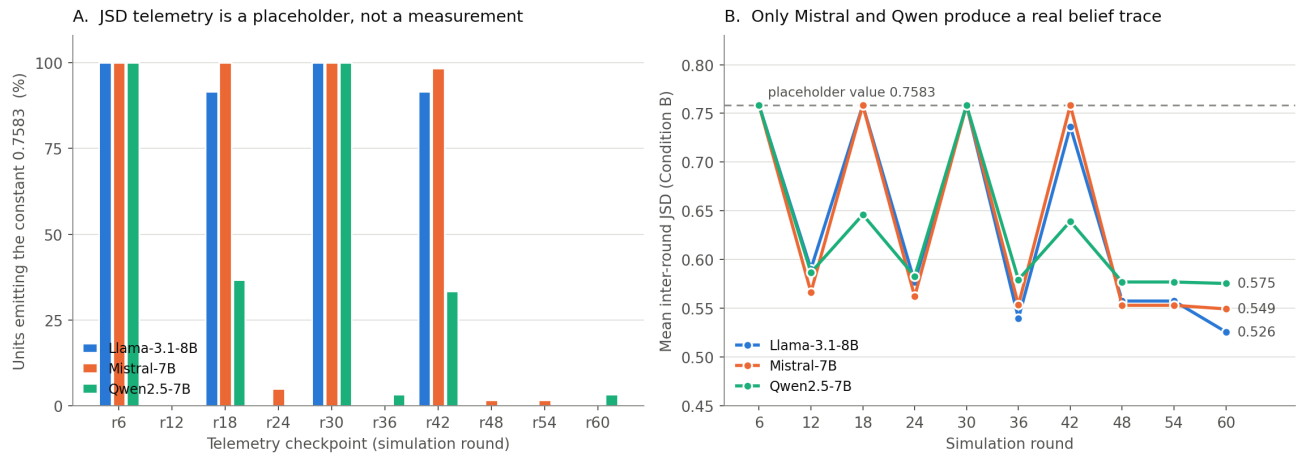


Fig. 17. Documentary reproduction: telemetry integrity diagnostic. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figA_jsd_integrity.png`.

Difficulty–lift coupling

Historical artifact 16/30; not a current result. The axes share the baseline score algebraically. A large correlation therefore does not validate a difficulty-based simulation-allocation rule. SBE/SDE classifications were selected after observing scores.

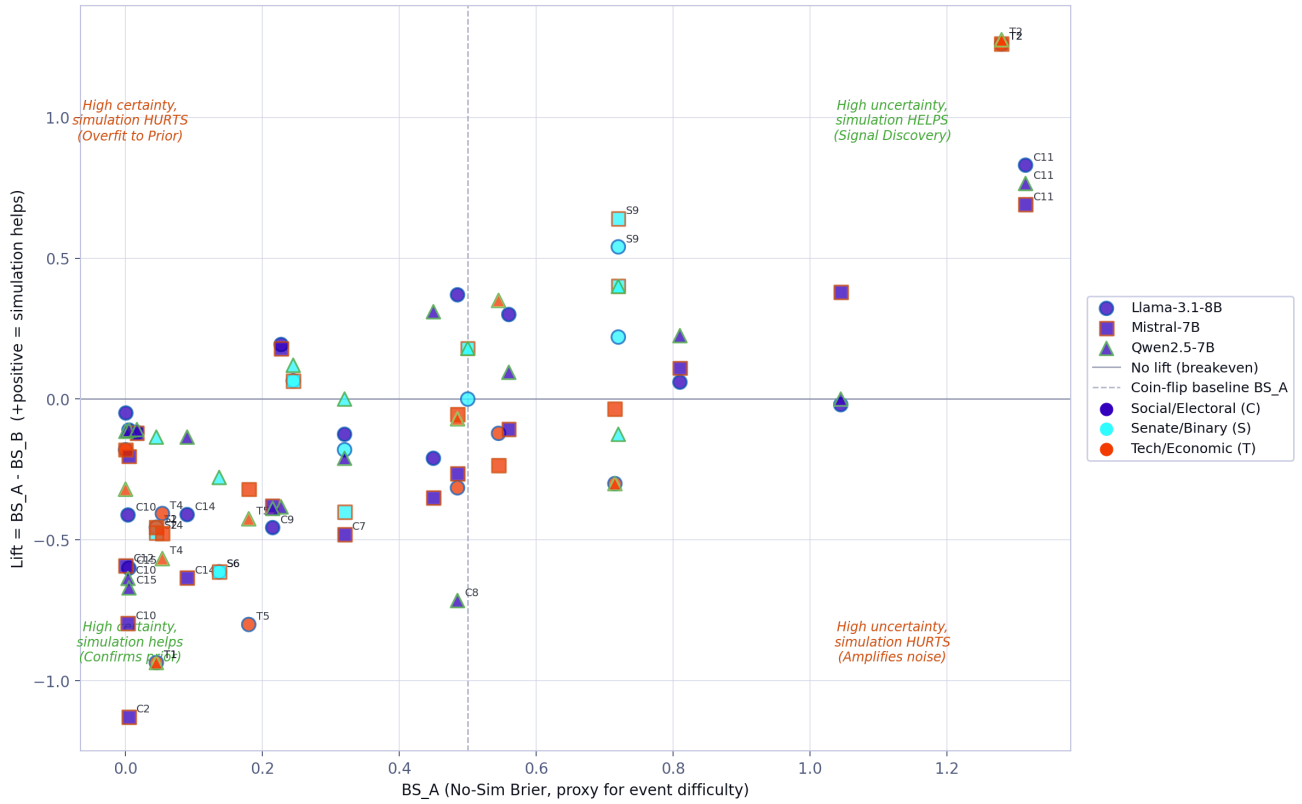


Fig. 18. Documentary reproduction: difficulty–lift coupling. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig6_volatility_vs_lift.png`.

Removing the shared term

Historical artifact 17/30; not a current result. The contrast illustrates mathematical coupling. A weak remaining correlation in a small selected dataset is not proof that difficulty has no predictive value; the plot does not estimate a causal mechanism.

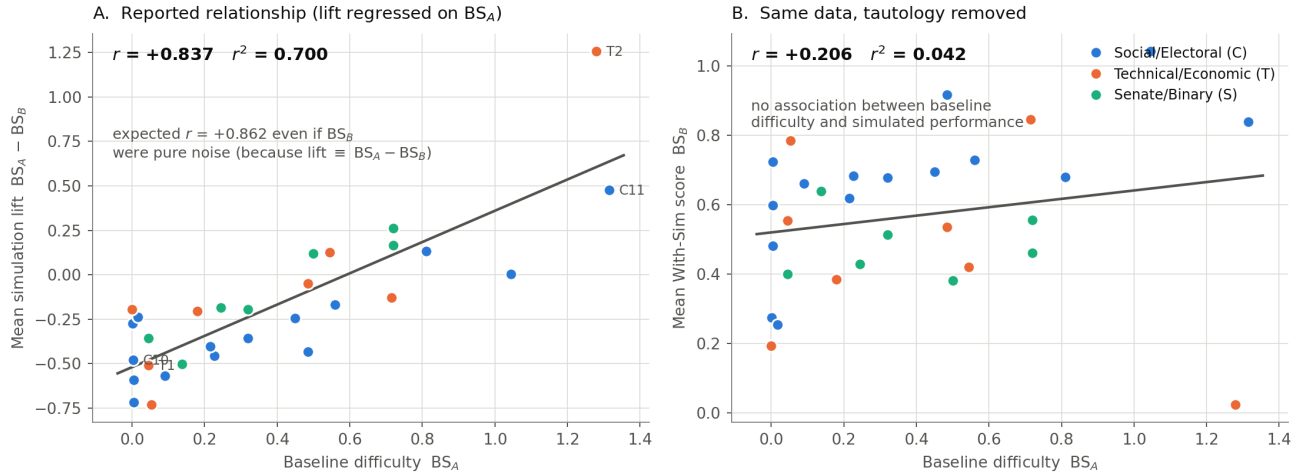


Fig. 19. Documentary reproduction: removing the shared term. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figD_difficulty_tautology.png`.

Superseded convergence profiles

Historical artifact 18/30; not a current result. This panel is expressly withdrawn as evidence of consensus. Placeholder-rich series and unclear reference distributions prevent the original interpretation. It is retained to document what the historical pipeline displayed.

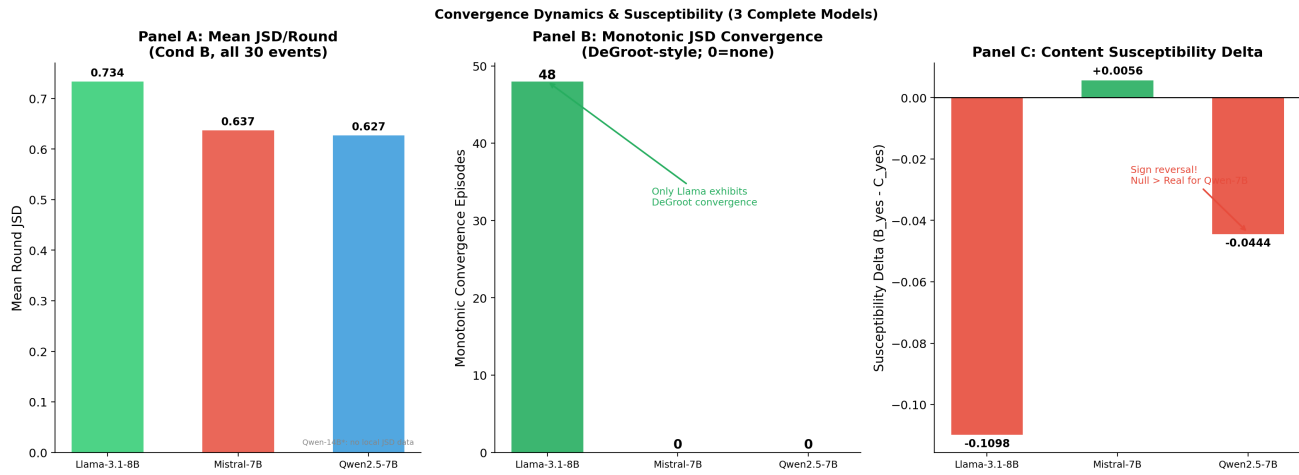


Fig. 20. Documentary reproduction: superseded convergence profiles. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig5_jsd_convergence.png`.

Historical directional outcomes

Historical artifact 19/30; not a current result. Agreement across campaigns can reflect shared inputs, knowledge or extraction choices. It does not establish that the social process amplifies a shared prior. Regenerated per-event probabilities determine current tie-aware accuracy.

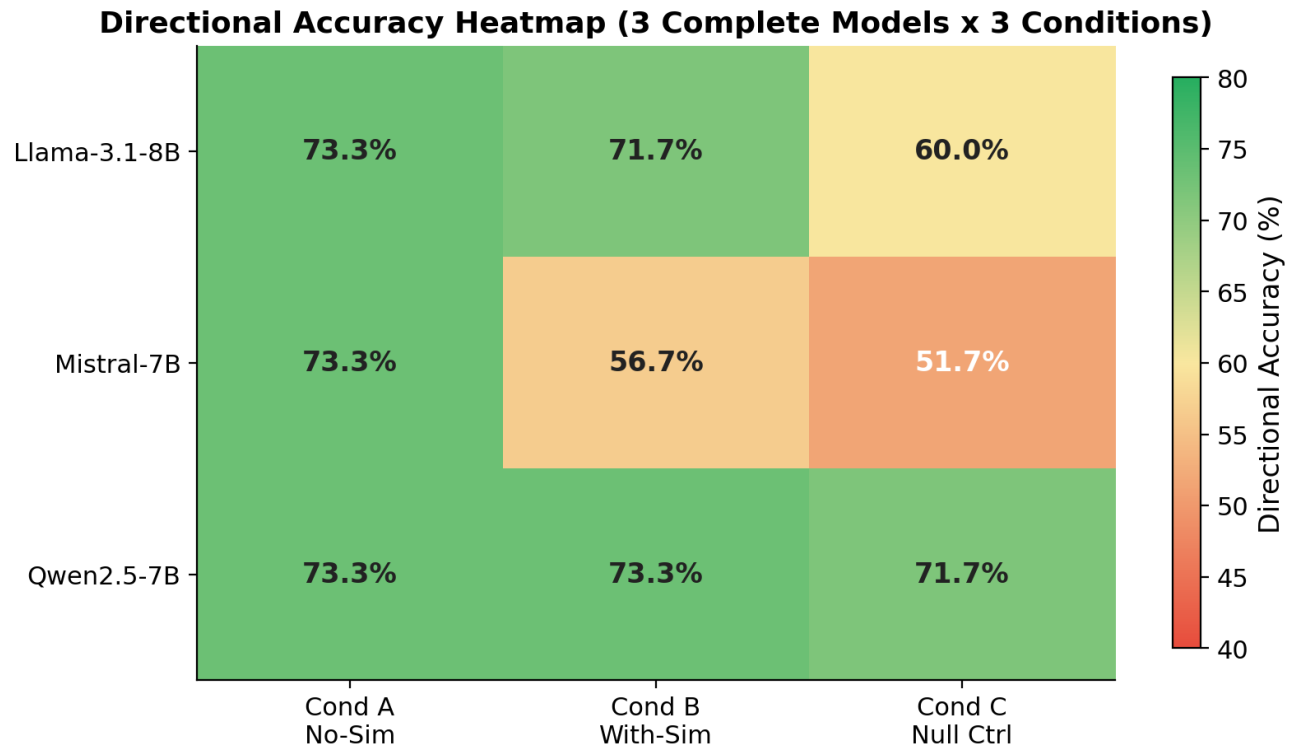


Fig. 21. Documentary reproduction: historical directional outcomes. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig7_directional_matrix.png`.

Score-threshold contamination heuristic

Historical artifact 20/30; not a current result. Low baseline Brier is not a test of training contamination. Thresholding a component of lift changes the selected lift distribution by construction. Labels claiming clean or contaminated are historical heuristic labels, not established classifications.

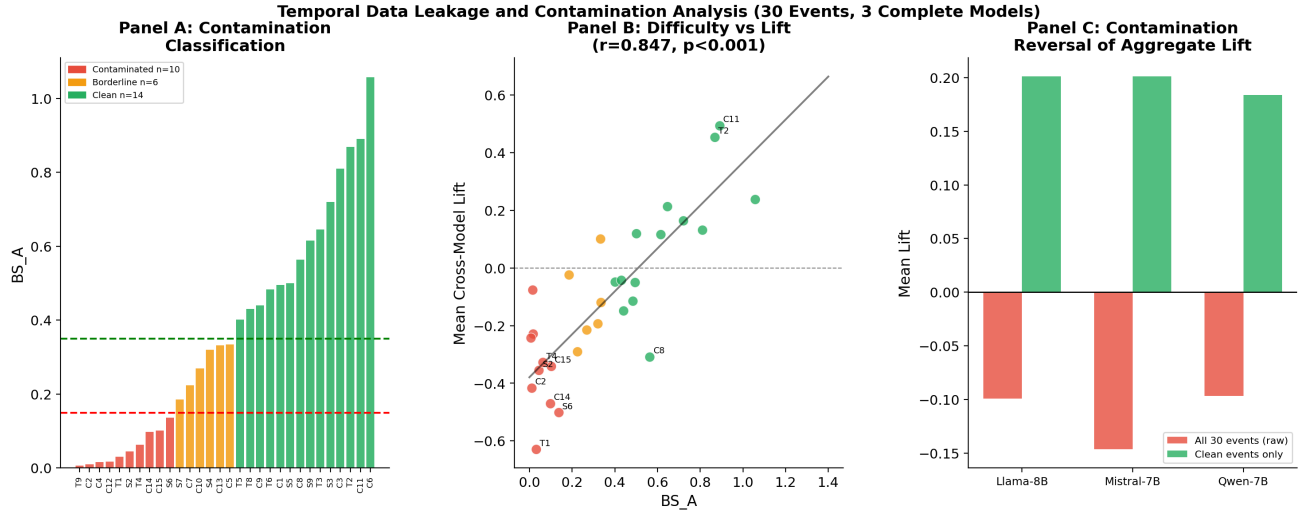


Fig. 22. Documentary reproduction: score-threshold contamination heuristic. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig9_contamination_analysis.png`.

Outcome-selected lift strata

Historical artifact 21/30; not a current result. Intervals do not remove bias from selecting strata on a measured score. A lift reversal after filtering low baseline loss is not evidence that leakage has been removed or simulation becomes beneficial.

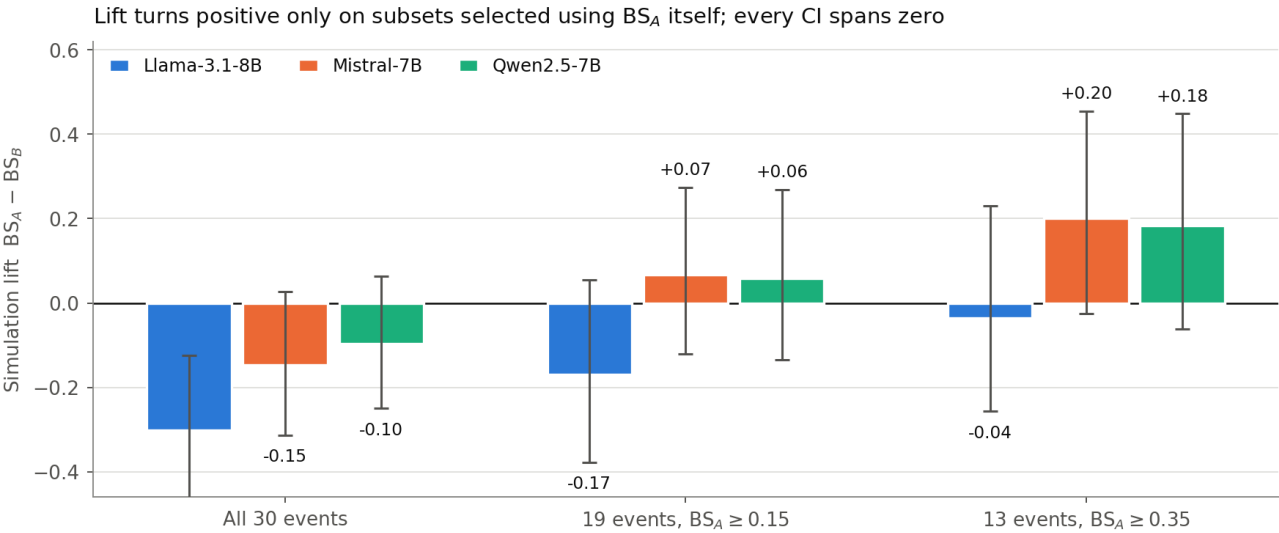


Fig. 23. Documentary reproduction: outcome-selected lift strata. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figC_lift_stratified.png`.

Release-date stratification

Historical artifact 22/30; not a current result. Release-date strata do not protect against later evaluator knowledge or hindsight in dossiers. Historical readings mix serving/evaluator states. The main paper preserves the 20-event selection only as a descriptive subset with sensitivity checks.

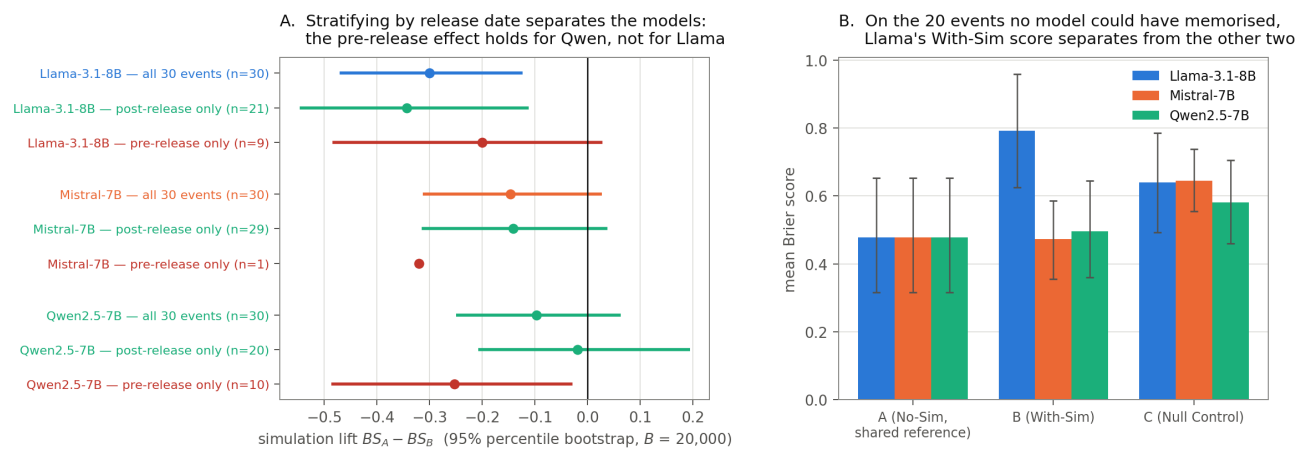


Fig. 24. Documentary reproduction: release-date stratification. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figH_cutoff_lift.png`.

Uniform-reference skill scores

Historical artifact 23/30; not a current result. BSS must state its normalization and aggregation convention. Uniform probability is not a competitive task-aware reference, and model ordering under one historical extractor is not a controlled capability ranking.

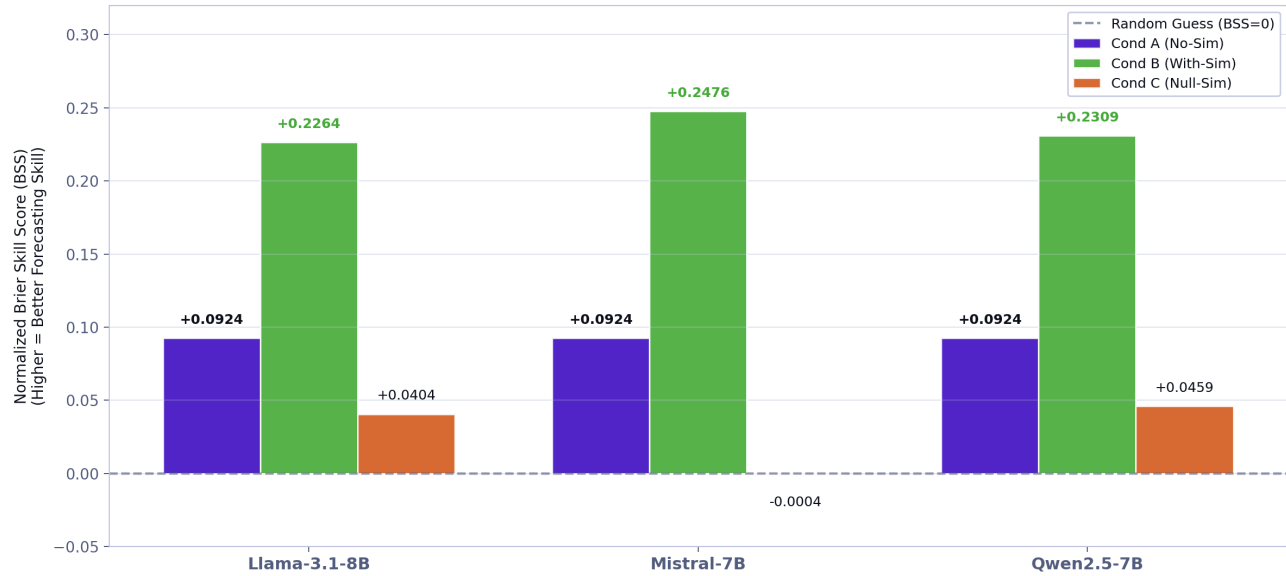


Fig. 25. Documentary reproduction: uniform-reference skill scores. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig10_bss_comparison.png`.

Binary and multicategory formats

Historical artifact 24/30; not a current result. Different option counts imply different uniform Brier references. Normalization alone does not equalize event difficulty or validate a binary-versus-MCQ generalization. Sample sizes and event selection remain limiting.

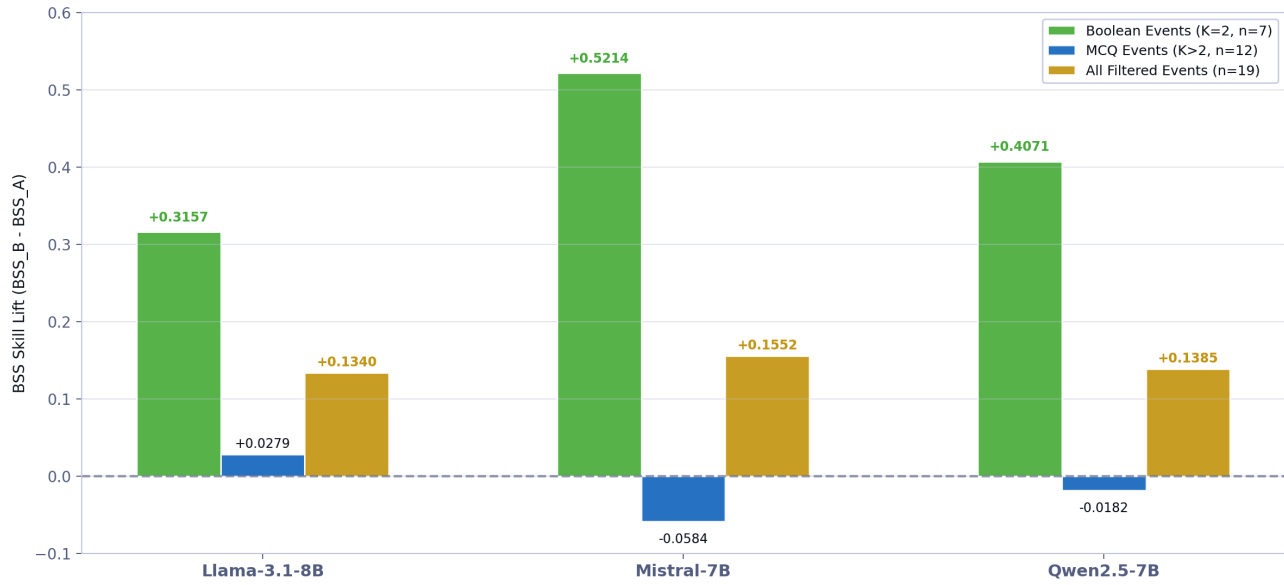


Fig. 26. Documentary reproduction: binary and multicategory formats. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `fig11_bss_by_format.png`.

Original Llama execution break

Historical artifact 25/30; not a current result. This is a pre-repair operational diagnostic. Completion flags can coexist with endpoint failures, repeated persona templates and absent telemetry. The current paired audit keeps only C3 and S2–S6 as untouched Llama evidence.

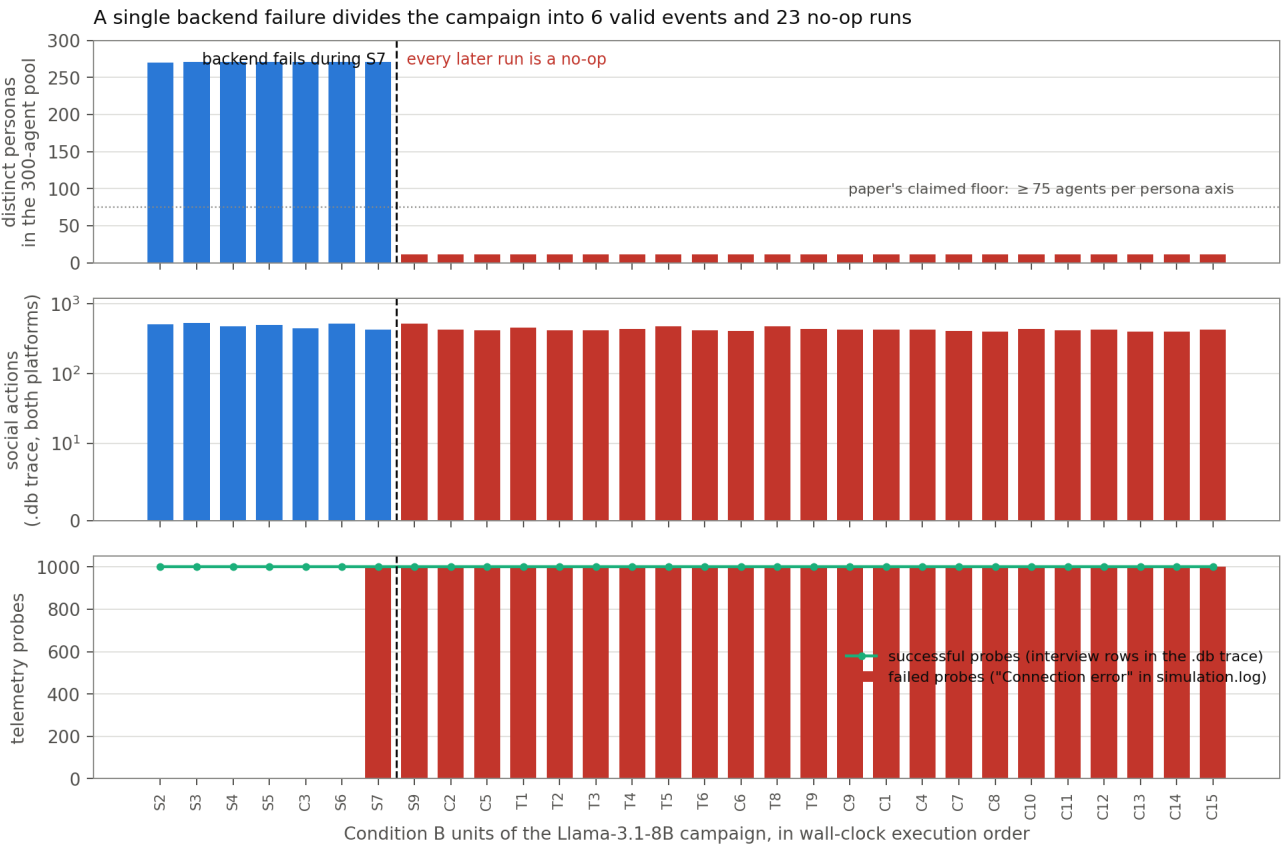


Fig. 27. Documentary reproduction: original llama execution break. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figF_llama_outage.png`.

Contents of the evaluator digest

Historical artifact 26/30; not a current result. Speaker counts and distinct texts concern a selected digest, not all conversation. Probe placeholders can populate sparse samples. Similar rubric scores on healthy and error-filled inputs warn about validity but do not identify the cause of each score.

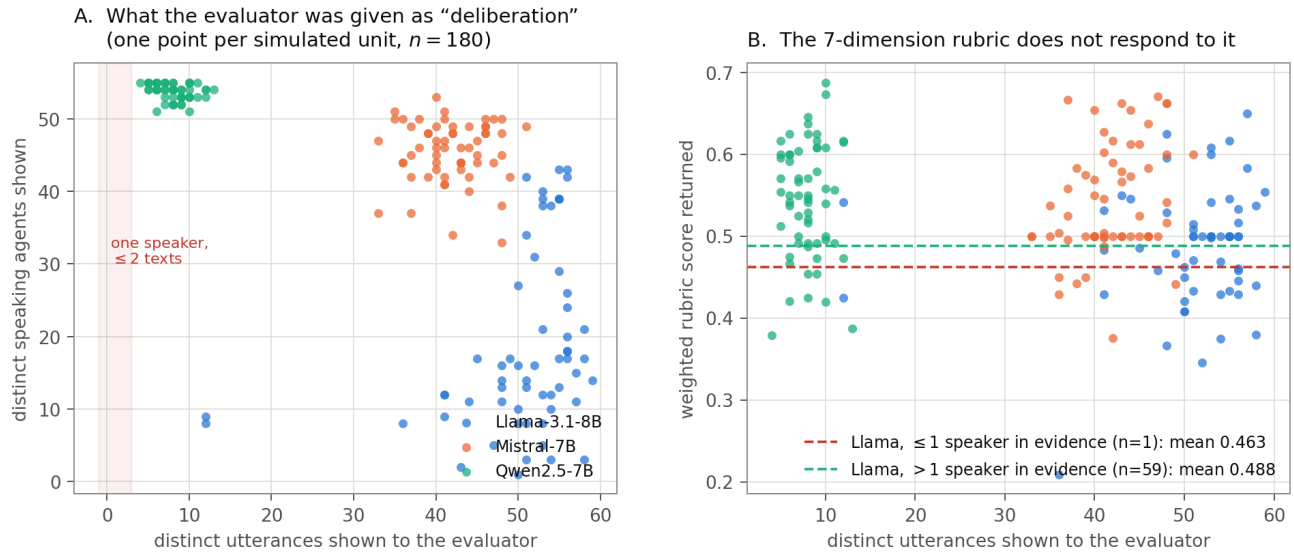


Fig. 28. Documentary reproduction: contents of the evaluator digest. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `figG_evidence_depth.png`.

Population sweep

Historical artifact 27/30; not a current result. The highlighted optimal zone is a withdrawn interpretation. C1's best observed B score is at 100 agents and C9's at 30. Single runs on two selected anchors do not establish a scaling law or agent-level precision guarantee.

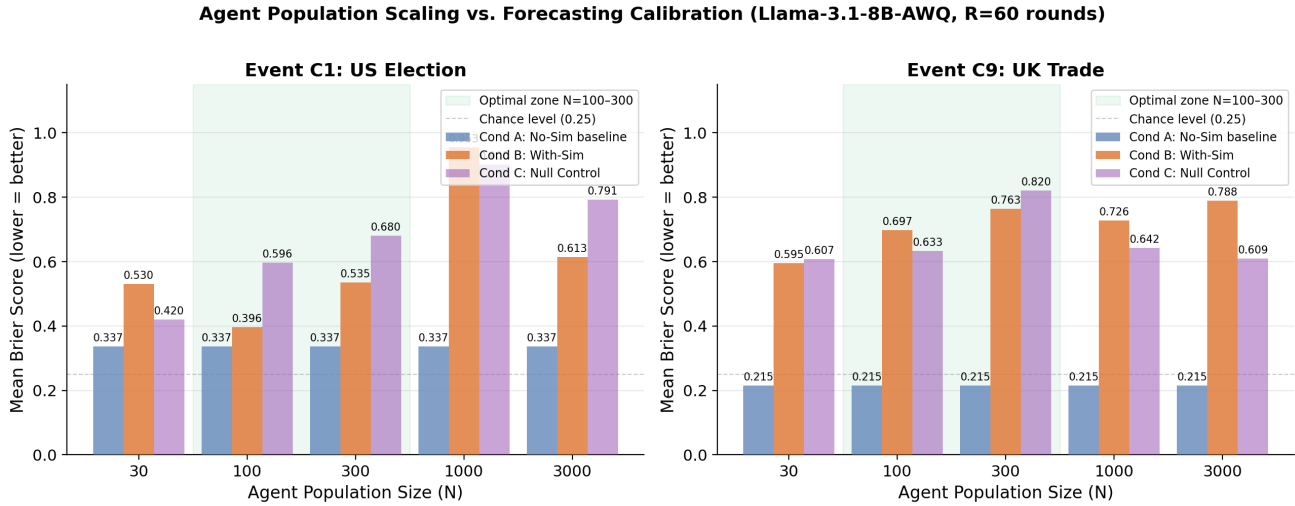


Fig. 29. Documentary reproduction: population sweep. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `n_sweep_calibration.png`.

Round sweep

Historical artifact 28/30; not a current result. The 60-round setting scored below 30 and 90 on these two recorded anchors. That does not locate a convergence time or echo-chamber threshold, and run-to-run uncertainty was not measured.

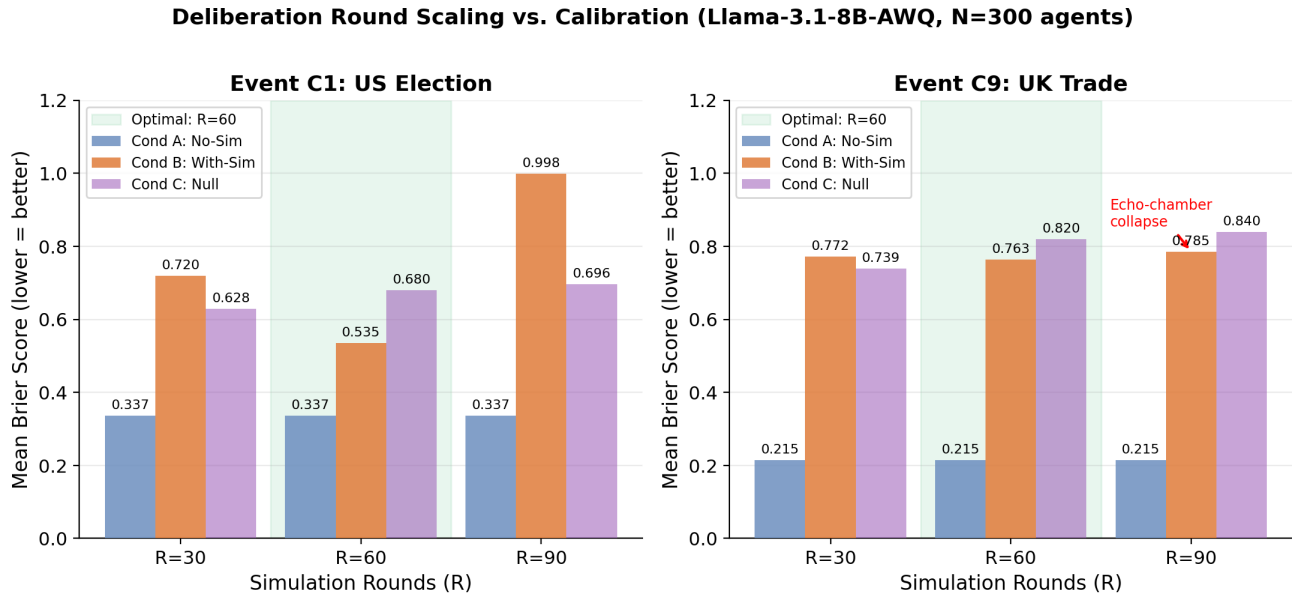


Fig. 30. Documentary reproduction: round sweep. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `r_sweep_calibration.png`.

Anchor-event cross-model comparison

Historical artifact 29/30; not a current result. Reference construction, model identity, serving, precision and coverage differ. These cells are historical engineering observations, not validation of a model-agnostic optimum.

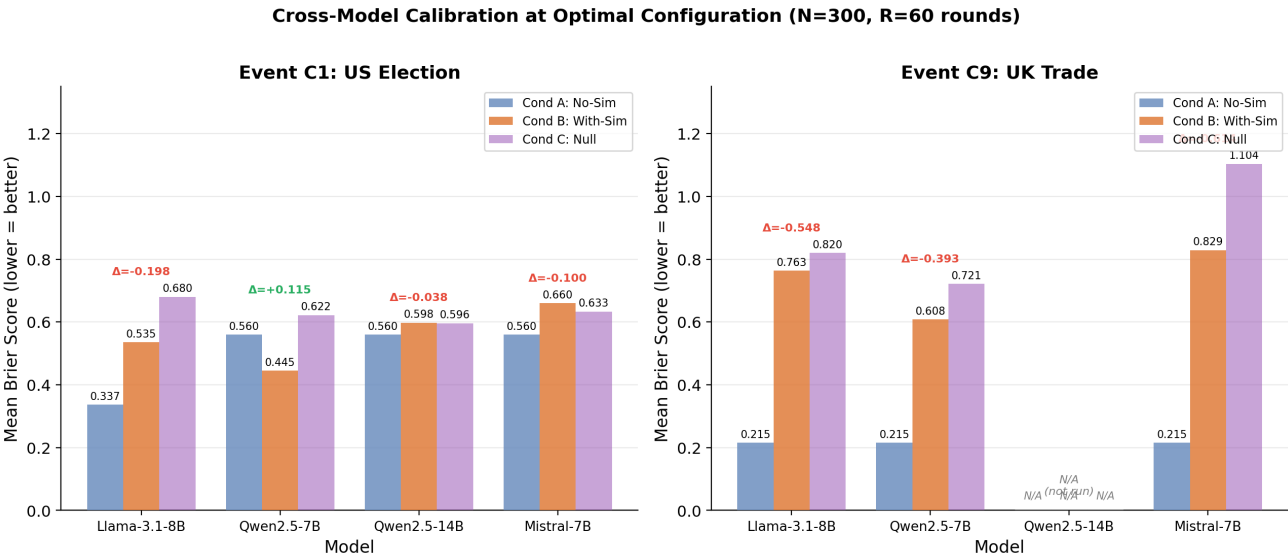


Fig. 31. Documentary reproduction: anchor-event cross-model comparison. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `cross_model_comparison.png`.

Historical computational-cost plot

Historical artifact 30/30; not a current result. Elapsed time is deployment-specific and not automatically GPU-hours or price. The shaded optimal zone is not an established optimum. Cost and loss require an explicit utility objective before a configuration can be recommended.

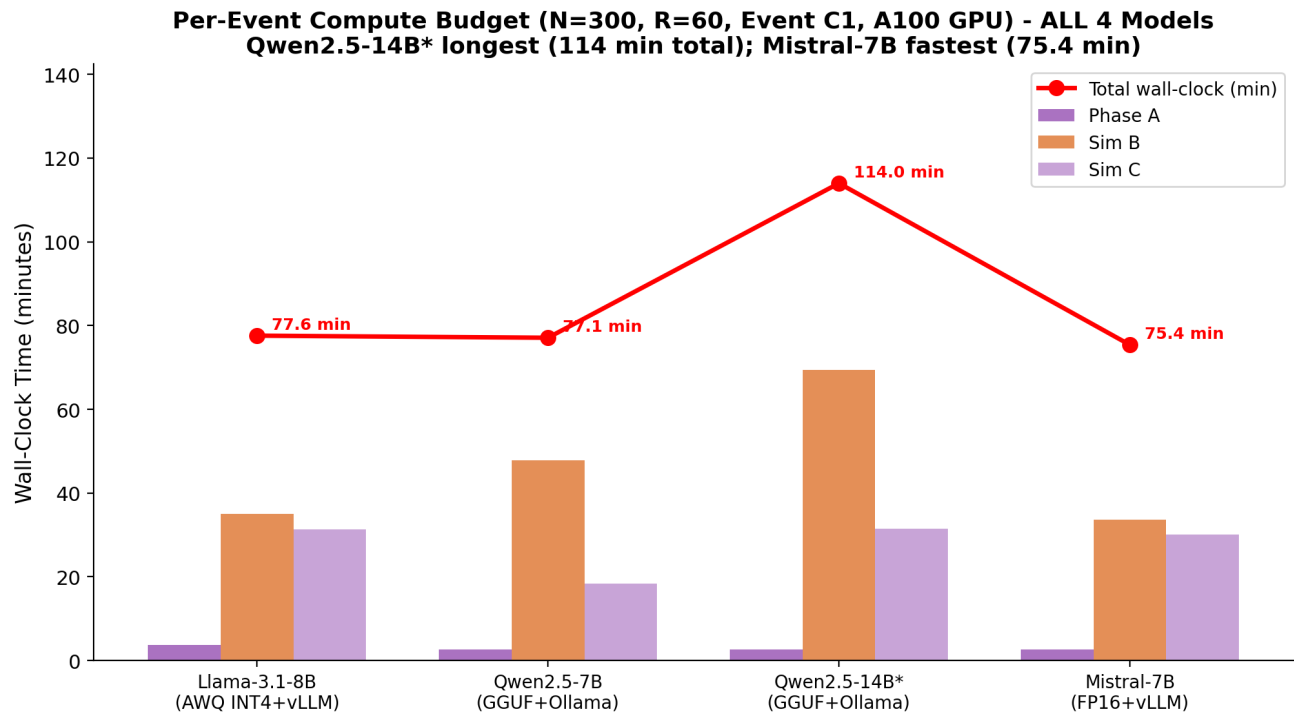


Fig. 32. Documentary reproduction: historical computational-cost plot. Embedded annotations are superseded where they conflict with the corrected interpretation above. Source file: `compute_cost_analysis.png`.

DATA AVAILABILITY

The replication package accompanies this manuscript and supports offline reanalysis. A permanent repository identifier has not yet been assigned in this working release; no public deposition or reviewer access is asserted. Scored records and evidence fields survive even where raw simulation traces were lost. Qwen2.5-14B’s original raw traces are unavailable for 21 events following scratch-storage deletion; recovery traces and the surviving original fragment are retained. The original evaluator deployment cannot be replayed exactly from the recorded artifacts. Reproducing the reported analysis and rerunning the historical system are distinct capabilities.

RESEARCH SCOPE

Inputs concern public events and were assembled for retrospective computational study. No new human-participant data were collected by the reanalysis. Synthetic personas and behaviours are not validated models of real populations. Results should not be interpreted as operational forecasting performance or as evidence about the beliefs of actual people.

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